US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication | 20250284099
Kind Code | A1
Publication Date | September 11, 2025
Inventor(s) | BAIK; Jae Hyun et al.

OPTICAL IMAGING SYSTEM

Abstract

An optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system, wherein the optical imaging system satisfies $1<|f134567-f|/f$, where f134567 is a composite focal length of the first to seventh lenses calculated with an index of refraction of the second lens set to 1.0, f is an overall focal length of the optical imaging system, and f134567 and f are expressed in a same unit of measurement.

Inventors: **BAIK; Jae Hyun (Suwon-si, KR), IM; Min Hyuk (Suwon-si, KR), JO; Yong Joo (Suwon-si, KR), AN; Ga Young (Suwon-si, KR)**

Applicant: **Samsung Electro-Mechanics Co., Ltd.** (Suwon-si, KR)

Family ID: **68694682**

Assignee: **Samsung Electro-Mechanics Co., Ltd. (Suwon-si, KR)**

Appl. No.: **19/219104**

Filed: **May 27, 2025**

Foreign Application Priority Data

KR	10-2018-0061393	May. 29, 2018
KR	10-2018-0106185	Sep. 05, 2018

Related U.S. Application Data

parent US continuation 18187874 20230322 parent-grant-document US 12345859 child US 19219104
parent US continuation 17088675 20201104 parent-grant-document US 11789239 child US 18187874

parent US continuation 16424774 20190529 parent-grant-document US 11353686 child US 17088675

Publication Classification

Int. Cl.: **G02B13/00** (20060101); **G02B9/64** (20060101); **H04N23/55** (20230101)

U.S. Cl.:

CPC **G02B13/0045** (20130101); **G02B9/64** (20130101); H04N23/55 (20230101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/187,874 filed on Mar. 22, 2023, which is a continuation of U.S. patent application Ser. No. 17/088,675 filed on Nov. 4, 2020, now U.S. Pat. No. 11,789,239 issued on Oct. 17, 2023, which is a continuation of U.S. patent application Ser. No. 16/424,774 filed on May 29, 2019, now U.S. Pat. No. 11,353,686 issued on Jun. 7, 2022, which claims the benefit under 35 USC 119 (a) of Korean Patent Application Nos. 10-2018-0061393 filed on May 29, 2018, and 10-2018-0106185 filed on Sep. 5, 2018, in the Korean Intellectual Property Office, the entire disclosures of which are incorporated herein by reference for all purposes.

BACKGROUND
1. Field
[0002] This application relates to an optical imaging system including seven lenses.
2. Description of Related Art
[0003] A mobile terminal is commonly provided with a camera for video communications or capturing images. However, it is difficult to achieve high performance in such a camera for a mobile terminal due to space limitations inside the mobile terminal.
[0004] Accordingly, a demand for an optical imaging system capable of improving the performance of the camera without increasing a size of the camera has increased as a number of mobile terminals provided with a camera has increased.
SUMMARY
[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.
[0006] In one general aspect, an optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system, wherein the optical imaging system satisfies $1<|f134567-f|/f$, where f134567 is a composite focal length of the first to seventh lenses calculated with an index of refraction of the second lens set to 1.0, f is an overall focal length of the optical imaging system, and f134567 and f are expressed in a same unit of measurement.
[0007] An object-side surface of the first lens may be convex.
[0008] An image-side surface of the seventh lens may be concave.
[0009] At least one inflection point may be formed on either one or both of an object-side surface and an image-side surface of the sixth lens.
[0010] At least one inflection point may be formed on either one or both of an object-side surface

and an image-side surface of the seventh lens.
[0011] A distance along the optical axis from an object-side surface of the first lens to the imaging plane may be 6.0 mm or less.
[0012] An F No. of the optical imaging system may be less than 1.7.
[0013] An object-side surface of the second lens may be convex.
[0014] An image-side surface of the third lens may be concave.
[0015] An object-side surface or an image-side surface of the fourth lens may be concave.
[0016] An image-side surface of the fifth lens may be concave.
[0017] Either one or both of an object-side surface and an image-side surface of the sixth lens may be convex.
[0018] The optical imaging system may further satisfy $0.1<L1w/L7w<0.3$, where L1w is a weight of the first lens, L7w is a weight of the seventh lens, and L1w and L7w are expressed in a same unit of measurement.
[0019] The optical imaging system may further include a spacer disposed between the sixth and seventh lenses, and the optical imaging system may further satisfy $0.5<S6d/f<1.2$, where S6d is an inner diameter of the spacer, f is the overall focal length of the optical imaging system, and S6d and f are expressed in a same unit of measurement.
[0020] The optical imaging system may further satisfy $0.4<L1TR/L7TR<0.7$, where L1TR is an overall outer diameter of the first lens, L7TR is an overall outer diameter of the seventh lens, and L1TR and L7TR are expressed in a same unit of measurement.
[0021] The optical imaging system may further satisfy $0.5<L1234TRavg/L7TR<0.75$, where L1234TRavg is an average value of overall outer diameters of the first to fourth lenses, L7TR is an overall diameter of the seventh lens, and L1234TRavg and L7TR are expressed in a same unit of measurement.
[0022] The optical imaging system may further satisfy $0.5<L12345TRavg/L7TR<0.76$, where L12345TRavg is an average value of overall outer diameters of the first to fifth lenses, L7TR is an overall outer diameter of the seventh lens, and L12345TRavg and L7TR are expressed in a same unit of measurement.
[0023] The second lens may have a positive refractive power.
[0024] The third lens may have a positive refractive power.
[0025] A paraxial region of an object-side surface of the seventh lens may be concave.
[0026] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0027] FIG. **1** is a view illustrating a first example of an optical imaging system.
[0028] FIG. **2** illustrates aberration curves of the optical imaging system of FIG. **1**.
[0029] FIG. **3** is a view illustrating a second example of an optical imaging system.
[0030] FIG. **4** illustrates aberration curves of the optical imaging system of FIG. **3**.
[0031] FIG. **5** is a view illustrating a third example of an optical imaging system.
[0032] FIG. **6** illustrates aberration curves of the optical imaging system of FIG. **5**.
[0033] FIG. **7** is a view illustrating a fourth example of an optical imaging system.
[0034] FIG. **8** illustrates aberration curves of the optical imaging system of FIG. **7**.
[0035] FIG. **9** is a view illustrating a fifth example of an optical imaging system.
[0036] FIG. **10** illustrates aberration curves of the optical imaging system of FIG. **9**.
[0037] FIG. **11** is a view illustrating a sixth example of an optical imaging system.
[0038] FIG. **12** illustrates aberration curves of the optical imaging system of FIG. **11**.

[0039] FIG. **13** is a view illustrating a seventh example of an optical imaging system.
[0040] FIG. **14** illustrates aberration curves of the optical imaging system of FIG. **13**.
[0041] FIG. **15** is a view illustrating an eighth example of an optical imaging system.
[0042] FIG. **16** illustrates aberration curves of the optical imaging system of FIG. **15**.
[0043] FIG. **17** is a view illustrating a ninth example of an optical imaging system.
[0044] FIG. **18** illustrates aberration curves of the optical imaging system of FIG. **17**.
[0045] FIG. **19** is a view illustrating a tenth example of an optical imaging system.
[0046] FIG. **20** illustrates aberration curves of the optical imaging system of FIG. **19**.
[0047] FIG. **21** is a view illustrating an eleventh example of an optical imaging system.
[0048] FIG. **22** illustrates aberration curves of the optical imaging system of FIG. **21**.
[0049] FIG. **23** is a view illustrating a twelfth example of an optical imaging system.
[0050] FIG. **24** illustrates aberration curves of the optical imaging system of FIG. **23**.
[0051] FIG. **25** is a view illustrating a thirteenth example of an optical imaging system.
[0052] FIG. **26** illustrates aberration curves of the optical imaging system of FIG. **25**.
[0053] FIG. **27** is a view illustrating a fourteenth example of an optical imaging system.
[0054] FIG. **28** illustrates aberration curves of the optical imaging system of FIG. **27**.
[0055] FIG. **29** is a view illustrating a fifteenth example of an optical imaging system.
[0056] FIG. **30** illustrates aberration curves of the optical imaging system of FIG. **29**.
[0057] FIG. **31** is a view illustrating a sixteenth example of an optical imaging system.
[0058] FIG. **32** illustrates aberration curves of the optical imaging system of FIG. **31**.
[0059] FIG. **33** is a view illustrating a seventeenth example of an optical imaging system.
[0060] FIG. **34** illustrates aberration curves of the optical imaging system of FIG. **33**.
[0061] FIG. **35** is a view illustrating an eighteenth example of an optical imaging system.
[0062] FIG. **36** illustrates aberration curves of the optical imaging system of FIG. **35**.
[0063] FIG. **37** is a view illustrating a nineteenth example of an optical imaging system.
[0064] FIG. **38** illustrates aberration curves of the optical imaging system of FIG. **37**.
[0065] FIG. **39** is a view illustrating a twentieth example of an optical imaging system.
[0066] FIG. **40** illustrates aberration curves of the optical imaging system of FIG. **39**.
[0067] FIG. **41** is a view illustrating a twenty-first example of an optical imaging system.
[0068] FIG. **42** illustrates aberration curves of the optical imaging system of FIG. **41**.
[0069] FIG. **43** is a view illustrating a twenty-second example of an optical imaging system.
[0070] FIG. **44** illustrates aberration curves of the optical imaging system of FIG. **43**.
[0071] FIG. **45** is a view illustrating a twenty-third example of an optical imaging system.
[0072] FIG. **46** illustrates aberration curves of the optical imaging system of FIG. **45**.
[0073] FIG. **47** is a view illustrating a twenty-fourth example of an optical imaging system.
[0074] FIG. **48** illustrates aberration curves of the optical imaging system of FIG. **47**.
[0075] FIG. **49** is a view illustrating a twenty-fifth example of an optical imaging system.
[0076] FIG. **50** illustrates aberration curves of the optical imaging system of FIG. **49**.
[0077] FIG. **51** is a view illustrating a twenty-sixth example of an optical imaging system.
[0078] FIG. **52** illustrates aberration curves of the optical imaging system of FIG. **51**.
[0079] FIG. **53** is a view illustrating a twenty-seventh example of an optical imaging system.
[0080] FIG. **54** illustrates aberration curves of the optical imaging system of FIG. **53**.
[0081] FIG. **55** is a view illustrating a twenty-eighth example of an optical imaging system.
[0082] FIG. **56** illustrates aberration curves of the optical imaging system of FIG. **55**.
[0083] FIGS. **57** and **58** are cross-sectional views illustrating examples of an optical imaging system and a lens barrel coupled to each other.
[0084] FIG. **59** is a cross-sectional view illustrating an example of a seventh lens.
[0085] FIG. **60** is a cross-sectional view illustrating an example of a shape of a rib of a lens.
[0086] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and

depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0087] The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order. Also, descriptions of features that are known in the art may be omitted for increased clarity and conciseness.

[0088] The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.

[0089] Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.

[0090] As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items.

[0091] Although terms such as “first,” “second,” and “third” may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

[0092] Spatially relative terms such as “above,” “upper,” “below,” and “lower” may be used herein for ease of description to describe one element's relationship to another element as shown in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being “above” or “upper” relative to another element will then be “below” or “lower” relative to the other element. Thus, the term “above” encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated by 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

[0093] The terminology used herein is for describing various examples only, and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes,” and “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

[0094] Thicknesses, sizes, and shapes of lenses illustrated in the drawings may have been slightly exaggerated for convenience of explanation. In addition, the shapes of spherical surfaces or aspherical surfaces of the lenses described in the detailed description and illustrated in the drawings are merely examples. That is, the shapes of the spherical surfaces or the aspherical surfaces of the

lenses are not limited to the examples described herein.
[0095] Numerical values of radii of curvature of lenses, thicknesses of lenses, distances between elements including lenses or surfaces, effective aperture radii of lenses, focal lengths, and diameters, thicknesses, and lengths of various elements are expressed in millimeters (mm), and angles are expressed in degrees. Thicknesses of lenses and distances between elements including lenses or surfaces are measured along the optical axis of the optical imaging system.
[0096] The term “effective aperture radius” as used in this application refers to a radius of a portion of a surface of a lens or other element (an object-side surface or an image-side surface of a lens or other element) through which light actually passes. The effective aperture radius is equal to a distance measured perpendicular to an optical axis of the surface between the optical axis of the surface and the outermost point on the surface through which light actually passes. Therefore, the effective aperture radius may be equal to a radius of an optical portion of a surface, or may be smaller than the radius of the optical portion of the surface if light does not pass through a peripheral portion of the optical portion of the surface. The object-side surface and the image-side surface of a lens or other element may have different effective aperture radii.
[0097] In this application, unless stated otherwise, a reference to the shape of a lens surface means the shape of a paraxial region of the lens surface. A paraxial region of a lens surface is a central portion of the lens surface surrounding the optical axis of the lens surface in which light rays incident to the lens surface make a small angle θ to the optical axis and the approximations $\sin\theta \approx \theta$, $\tan\theta \approx \theta$, and $\cos\theta \approx 1$ are valid.
[0098] For example, a statement that the object-side surface of a lens is convex means that at least a paraxial region of the object-side surface of the lens is convex, and a statement that the image-side surface of the lens is concave means that at least a paraxial region of the image-side surface of the lens is concave. Therefore, even though the object-side surface of the lens may be described as convex, the entire object-side surface of the lens may not be convex, and a peripheral region of the object-side surface of the lens may be concave. Also, even though the image-side surface of the lens may be described as concave, the entire image-side surface of the lens may not be concave, and a peripheral region of the image-side surface of the lens may be convex.
[0099] An optical imaging system includes a plurality of lenses. For example, the optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system. Thus, the first lens is a lens closest to an object (or a subject) to be imaged by the optical imaging system, while the seventh lens is a lens closest to the imaging plane.
[0100] Each lens of the optical imaging system includes an optical portion and a rib. The optical portion of the lens is a portion of the lens that is configured to refract light, and is generally formed in a central portion of the lens. The rib of the lens is an edge portion of the lens that enables the lens to be mounted in a lens barrel and the optical axis of the lens to be aligned with the optical axis of the optical imaging system. The rib of the lens extends radially outward from the optical portion. The optical portions of the lenses are generally not in contact with each other. For example, the first to seventh lenses are mounted in the lens barrel so that they are spaced apart from one other by predetermined distances along the optical axis of the optical imaging system. The ribs of the lenses may be selectively in contact with each other. For example, the ribs of the first to fourth lenses, or the first to fifth lenses, or the second to fourth lenses, may be in contact with each other so that the optical axes of these lenses may be easily aligned with the optical axis of the optical imaging system.
[0101] Next, a configuration of the optical imaging system will be described.
[0102] The optical imaging system includes a plurality of lenses. For example, the optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging

system from an object side of the optical imaging system toward an imaging plane of the optical imaging system.
[0103] The optical imaging system further includes an image sensor and a filter. The image sensor forms an imaging plane and converts light refracted by the first to seventh lenses into an electrical signal. The filter is disposed between the seventh lens and the imaging plane, and blocks infrared rays in the light refracted by the first to seventh lenses from being incident on the imaging plane.
[0104] The optical imaging system further includes a stop and spacers. The stop may be disposed in front of the first lens, or at a position of either an object-side surface or an image side-surface of one of the first to seventh lenses, or between two adjacent lenses of the first to seventh lenses, or between the object-side surface and the image-side surface of one of the first to seventh lenses, to adjust an amount of light incident on the imaging plane. Some examples may include two stops, one of which may be disposed in front of the first lens, or at the position of the object-side surface of the first lens, or between the object-side surface and the image-side surface of the first lens. Each of the spacers is disposed at a respective position between two lenses of the first to seventh lenses to maintain a predetermined distance between the two lenses. In addition, the spacers may be formed of a light-shielding material to block extraneous light from penetrating into the ribs of the lenses. There may be six or seven spacers. For example, a first spacer is disposed between the first lens and the second lens, a second spacer is disposed between the second lens and the third lens, a third spacer is disposed between the third lens and the fourth lens, a fourth spacer is disposed between the fourth lens and the fifth lens, a fifth spacer is disposed between the fifth lens and the sixth lens, and a sixth spacer is disposed between the sixth lens and the seventh lens. In addition, the optical imaging system may further include a seventh spacer disposed between the sixth lens and the sixth spacer.
[0105] Next, the lenses of the optical imaging system will be described.
[0106] The first lens has a refractive power. For example, the first lens has a positive refractive power or a negative refractive power. One surface of the first lens may be convex. For example, an object-side surface of the first lens may be convex. The first lens may include an aspherical surface. For example, one surface or both surfaces of the first lens may be aspherical.
[0107] The second lens has a refractive power. For example, the second lens has a positive refractive power or a negative refractive power. One surface or both surfaces of the second lens may be convex. For example, an object-side surface or both an object-side surface and an image-side surface of the second lens may be convex. The second lens may include an aspherical surface. For example, one surface or both surfaces of the second lens may be aspherical.
[0108] The third lens has a refractive power. For example, the third lens has a positive refractive power or a negative refractive power. One surface or both surfaces of the third lens may be convex. For example, an object-side surface, or an image-side surface, or both an object-side surface and an image-side surface of the third lens may be convex. The third lens may include an aspherical surface. For example, one surface or both surfaces of the third lens may be aspherical.
[0109] The fourth lens has a refractive power. For example, the fourth lens has a positive refractive power or a negative refractive power. One surface or both surfaces of the fourth lens may be convex. For example, an object-side surface, or an image-side surface, or both an object-side surface and an image-side surface of the fourth lens may be convex. The fourth lens may include an aspherical surface. For example, one surface or both surfaces of the fourth lens may be aspherical.
[0110] The fifth lens has a refractive power. For example, the fifth lens has a positive refractive power or a negative refractive power. One surface of the fifth lens may be convex. For example, an object-side surface or an image-side surface of the fifth lens may be convex. The fifth lens may include an aspherical surface. For example, one surface or both surfaces of the fifth lens may be aspherical.
[0111] The sixth lens has a refractive power. For example, the sixth lens has a positive refractive power or a negative refractive power. In some examples, one surface or both surfaces of the sixth

lens may be convex. For example, an object-side surface or both an object-side surface and an image-side surface of the sixth lens may be convex. Also, in some examples, one surface or both surfaces of the sixth lens may be concave. For example, an image-side surface or both an object-side surface and an image-side surface of the sixth lens may be concave. At least one surface of the sixth lens may have at least one inflection point. An inflection point is a point where a lens surface changes from convex to concave, or from concave to convex. A number of inflection points is counted from a center of the lens to an outer edge of the optical portion of the lens. For example, at least one inflection point may be formed on either one or both of the object-side surface and the image-side surface of the sixth lens. Therefore, at least one surface of the sixth lens may have a paraxial region and a peripheral region having shapes that are different from each other. For example, a paraxial region of the image-side surface of the sixth lens may be concave, but a peripheral region thereof may be convex. The sixth lens may have an aspherical surface. For example, one surface or both surfaces of the sixth lens may be aspherical.

[0112] The seventh lens has a refractive power. For example, the seventh lens has a positive refractive power or a negative refractive power. One surface of the seventh lens may be concave. For example, an image-side surface or both an object-side surface and an image-side surface of the seventh lens may be concave. At least one surface of the seventh lens may have at least one inflection point. For example, at least one inflection point may be formed on either one or both of the object-side surface and the image-side surface of the seventh lens. Therefore, at least one surface of the seventh lens may have a paraxial region and a peripheral region having shapes that are different from each other. For example, a paraxial region of the image-side surface of the seventh lens may be concave, but a peripheral region thereof may be convex. The seventh lens may include an aspherical surface. For example, one surface or both surfaces of the seventh lens may be aspherical.

[0113] The lenses of the optical imaging system may be made of a light material having a high light transmittance. For example, the first to seventh lenses may be made of a plastic material. However, a material of the first to seventh lenses is not limited to the plastic material.

[0114] The aspherical surfaces of the first to seventh lenses may be represented by the following Equation 1:

[00001]

$$Z = \frac{cY^2}{1+\sqrt{1-(1+K)c^2Y^2}} + AY^4 + BY^6 + CY^8 + DY^{10} + EY^{12} + FY^{14} + GY^{16} + HY^{18} + \ldots \quad (1)$$

.Math.

[0115] In Equation 1, c is a curvature of a lens surface and is equal to an inverse of a radius of curvature of the lens surface at an optical axis of the lens surface, K is a conic constant, Y is a distance from a certain point on an aspherical surface of the lens to an optical axis of the lens in a direction perpendicular to the optical axis, A to H are aspherical constants, and Z (or sag) is a distance between the certain point on the aspherical surface of the lens at the distance Y to the optical axis and a tangential plane perpendicular to the optical axis meeting the apex of the aspherical surface of the lens. Some of the examples disclosed in this application include an aspherical constant J. An additional term of JY.sup.20 may be added to the right side of Equation 1 to reflect the effect of the aspherical constant J.

[0116] The optical imaging system may satisfy one or more of the following Conditional Expressions 1 to 6:

[00002] $0.1 < L1w/L7w < 0.4 \quad \text{(ConditionalExpression1)}$

$0.5 < S6d/f < 1.4 \quad \text{(ConditionalExpression2)}$

$0.4 < L1\mathrm{TR}/L7\mathrm{TR} < 0.8 \quad \text{(ConditionalExpression3)}$

$0.5 < L1234\mathrm{TRavg}/L7\mathrm{TR} < 0.9 \quad \text{(ConditionalExpression4)}$

$0.5 < L12345\mathrm{TRavg}/L7\mathrm{TR} < 0.9 \quad \text{(ConditionalExpression5)}$

$1 < .\text{Math.} f134567 - f .\text{Math.} / f$ (ConditionalExpression6)

[0117] In the above Conditional Expressions, L1w is a weight of the first lens, and L7w is a weight of the seventh lens.

[0118] S6d is an inner diameter of the sixth spacer, and f is an overall focal length of the optical imaging system.

[0119] L1TR is an overall outer diameter of the first lens, and L7TR is an overall outer diameter of the seventh lens. The overall outer diameter of a lens is a diameter of the lens including both the optical portion of the lens and the rib of the lens.

[0120] L1234TRavg is an average value of overall outer diameters of the first to fourth lenses, and L12345TRavg is an average value of overall outer diameters of the first to fifth lenses.

[0121] f134567 is a composite focal length of the first to seventh lenses calculated with an index of refraction of the second lens set to 1.0, which is substantially equal to an index of refraction of air. When the index of refraction of the second lens is set to 1.0, the second lens does not refract light. Thus, by comparing f134567 with f, which is the overall focal length of the optical system, it is possible to evaluate the effect of the second lens on f. For example, the second lens may shorten f, or lengthen f, or have no effect on f In other words, f134567 may be greater than f, or less than f, or equal to f.

[0122] Conditional Expressions 1 and 3 specify ranges of a weight ratio and an overall outer diameter ratio between the first lens and the seventh lens to facilitate a self-alignment between the lenses and an alignment by a lens barrel.

[0123] Conditional Expression 2 specifies a range of a ratio of the inner diameter of the sixth spacer to the overall focal length of the optical imaging system for minimizing a flare phenomenon.

[0124] Conditional Expressions 4 and 5 specify overall outer diameter ratios between the lenses to facilitate aberration correction.

[0125] Conditional Expression 6 specifies a lower limit of a degree of shortening of f, which is the overall focal length of the optical imaging system, by the second lens. The lower limit of 1 for |f134567−f|/f in Conditional Expression 6 corresponds to an example in which the second lens shortens f to 50% of f134567. Thus, Conditional Expression 6 covers examples in which f is 50% or less of f134567.

[0126] The optical imaging system may also satisfy one or more of the following Conditional Expressions 7 to 12:

[00003] $0.1 < L1w / L7w < 0.3$ (ConditionalExpression7)

$0.5 < S6d / f < 1.2$ (ConditionalExpression8)

$0.4 < L1TR / L7TR < 0.7$ (ConditionalExpression9)

$0.5 < L1234TRavg / L7TR < 0.75$ (ConditionalExpression10)

$0.5 < L12345TRavg / L7TR < 0.76$ (ConditionalExpression11)

$1 < .\text{Math.} f134567f .\text{Math.} / f < 100$ (ConditionalExpression12) [0127] Conditional Expressions 7 to 12 are the same as Conditional Expressions 1 to 6, except that Conditional Expressions 7 to 12 specify narrower ranges.

[0128] The optical imaging system may also satisfy one or more of the following Conditional Expressions 13 to 33:

[00004] $0.01 < R1 / R4 < 1.3$ (ConditionalExpression13)

$0.1 < R1 / R5 < 0.7$ (ConditionalExpression14)

$0.05 < R1 / R6 < 0.9$ (ConditionalExpression15)

$0.2 < R1 / R11 < 1.2$ (ConditionalExpression16)

$0.8 < R1 / R14 < 1.2$ (ConditionalExpression17)

$0.6 < (R11 + R14) / (2 * R1) < 3.$ (ConditionalExpression18)

$0.4 < D13 / D57 < 1.2$ (ConditionalExpression19)

$0.1 < (1/f1 + 1/f2 + 1/f3 + 1/f4 + 1/f5 + 1/f6 + 1/f7) * f < 0.8$ (ConditionalExpression20)

$0.1 < (1/f1 + 1/f2 + 1/f3 + 1/f4 + 1/f5 + 1/f6 + 1/f7) * \mathrm{TTL} < 1.$ (ConditionalExpression21)

$0.2 < \mathrm{TD1} / D67 < 0.8$ (ConditionalExpression22)

$0.1 < (R11 + R14) / (R5 + R6) < 1.$ (ConditionalExpression23)

$\mathrm{SD12} < \mathrm{SD34}$ (ConditionalExpression24) $\mathrm{SD56} < \mathrm{SD67}$ (ConditionalExpression25)

$\mathrm{SD56} < \mathrm{SD34}$ (ConditionalExpression26)

$0.6 < \mathrm{TTL} / (2 * (\mathrm{IMGHT})) < 0.9$ (ConditionalExpression27)

0.2 < .Math. SD / .Math. TD < 0.7 (ConditionalExpression28)

$0 < \min(f1{:}f3) / \max(f4{:}f7) < 0.4$ (ConditionalExpression29)

0.4 < (.Math. TD) / TTL < 0.7 (ConditionalExpression30)

$0.7 < \mathrm{SL} / \mathrm{TTL} < 1.$ (ConditionalExpression31)

$0.81 < f12 / f123 < 0.96$ (ConditionalExpression32)

$0.6 < f12 / f1234 < 0.84$ (ConditionalExpression33)

[0129] In the above Conditional Expressions, R1 is a radius of curvature of an object-side surface of the first lens, R4 is a radius of curvature of an image-side surface of the second lens, R5 is a radius of curvature of an object-side surface of the third lens, R6 is a radius of curvature of an image-side surface of the third lens, R11 is a radius of curvature of an object-side surface of the sixth lens, and R14 is a radius of curvature of an image-side surface of the seventh lens.

[0130] D13 is a distance along an optical axis of the optical imaging system from the object-side surface of the first lens to the image-side surface of the third lens, and D57 is a distance along the optical axis from an object-side surface of the fifth lens to the image-side surface of the seventh lens.

[0131] f1 is a focal length of the first lens, f2 is a focal length of the second lens, f3 is a focal length of the third lens, f4 is a focal length of the fourth lens, f5 is a focal length of the fifth lens, f6 is a focal length of the sixth lens, f7 is a focal length of the seventh lens, f is an overall focal length of the optical imaging system, and TTL is a distance along the optical axis from the object-side surface of the first lens to an imaging plane of an image sensor of the optical imaging system.

[0132] TD1 is a thickness along the optical axis of the first lens, and D67 is a distance along the optical axis from the object-side surface of the sixth lens to the image-side surface of the seventh lens.

[0133] SD12 is a distance along the optical axis from an image-side surface of the first lens to an object-side surface of the second lens, SD34 is a distance along the optical axis from the image-side surface of the third lens to an object-side surface of the fourth lens, SD56 is a distance along the optical axis from an image-side surface of the fifth lens to the object-side surface of the sixth lens, and SD67 is a distance along the optical axis from an image-side surface of the sixth lens to an object-side surface of the seventh lens.

[0134] IMG HT is one-half of a diagonal length of the imaging plane of the image sensor.

[0135] ΣSD is a sum of air gaps along the optical axis between the lenses, and ΣTD is a sum of thicknesses along the optical axis of the lenses. An air gap is a distance along the optical axis between adjacent lenses.

[0136] min (f1:f3) is a minimum value of absolute values of the focal lengths of the first to third lenses, and max (f4:f7) is a maximum value of absolute values of the focal lengths of the fourth to seventh lenses.

[0137] SL is a distance along the optical axis from a stop to the imaging plane of the image sensor.

[0138] f12 is a composite focal length of the first and second lenses, f123 is a composite focal length of the first to third lenses, and f1234 is a composite focal length of the first to fourth lenses.

[0139] Conditional Expression 13 specifies a design range of the second lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the second lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 13, and it is difficult to expect a sufficient correction of astigmatic field curves for the second lens having a radius of curvature that is below the lower limit value of Conditional Expression 13.

[0140] Conditional Expressions 14 and 15 specify a design range of the third lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the third lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 14 or 15, and it is difficult to expect a correction of astigmatic field curves for the third lens having a radius of curvature that is below the lower limit value of Conditional Expression 14 or 15.

[0141] Conditional Expression 16 specifies a design range of the sixth lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the sixth lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 16, and the sixth lens having a radius of curvature that is below the lower limit value of Conditional Expression 16 is apt to cause a flare phenomenon.

[0142] Conditional Expression 17 specifies a design range of the seventh lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the seventh lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 17, and the seventh lens having a radius of curvature that is below the lower limit value of Conditional Expression 17 tends to cause a curvature of the imaging plane.

[0143] Conditional Expression 18 specifies a ratio of a sum of radii of curvature of the sixth lens and the seventh lens to twice a radius of curvature of the first lens for correcting the longitudinal spherical aberration and achieving excellent optical performance.

[0144] Conditional Expression 19 specifies a ratio of an optical imaging system mountable in a small terminal. For example, an optical imaging system having a ratio that exceeds the upper limit value of Conditional Expression 19 may cause a problem that the total length of the optical imaging system becomes long, and an optical imaging system having a ratio that is below the lower limit value of Conditional Expression 19 may cause a problem that a size of a lateral cross-section of the optical imaging system becomes large.

[0145] Conditional Expressions 20 and 21 specify a refractive power ratio of the first to seventh lenses for facilitating mass production of the optical imaging system. For example, an optical imaging system that exceeds the upper limit value of Conditional Expression 20 or 21 or is below the lower limit value of Conditional Expression 20 or 21 is difficult to commercialize because the refractive power of one or more of the first to seventh lenses is too great.

[0146] Conditional Expression 22 specifies a thickness range of the first lens for implementing a compact optical imaging system. For example, the first lens having a thickness that exceeds the upper limit value of Conditional Expression 22 or is below the lower limit value of Conditional Expression 22 is too thick or too thin to be manufactured.

[0147] Conditional Expression 24 specifies a design condition of the first to fourth lenses for improving chromatic aberration. For example, a case in which a distance between the first lens and the second lens is shorter than a distance between the third lens and the fourth lens is advantageous for improving the chromatic aberration.

[0148] Conditional Expressions 27 to 30 specify design conditions for implementing a compact optical imaging system. For example, lenses that deviate from the numerical range of Conditional

Expression 28 or 30 are difficult to form by injection molding and process.

[0149] Conditional Expressions 31 to 33 specify design conditions of an optical imaging system in consideration of a position of the stop. For example, an optical imaging system that does not satisfy one or more of Conditional Expressions 31 to 33 may have a longer overall length due to the refractive power of the lenses disposed behind the stop.

[0150] Next, various examples of the optical imaging system will be described. In the tables that appear in the following examples, S1 denotes an object-side surface of the first lens, S2 denotes an image-side surface of the first lens, S3 denotes an object-side surface of the second lens, S4 denotes an image-side surface of the second lens, S5 denotes an object-side surface of the third lens, S6 denotes an image-side surface of the third lens, S7 denotes an object-side surface of the fourth lens, S8 denotes an image-side surface of the fourth lens, S9 denotes an object-side surface of the fifth lens, S10 denotes an image-side surface of the fifth lens, S11 denotes an object-side surface of the sixth lens, S12 denotes an image-side surface of the sixth lens, S13 denotes an object-side surface of the seventh lens, S14 denotes an image-side surface of the seventh lens, S15 denotes an object-side surface of the filter, S16 denotes an image-side surface of the filter, and S17 denotes the imaging plane.

First Example

[0151] FIG. **1** is a view illustrating a first example of an optical imaging system, and FIG. **2** illustrates aberration curves of the optical imaging system of FIG. **1**.

[0152] An optical imaging system **1** includes a first lens **1001**, a second lens **2001**, a third lens **3001**, a fourth lens **4001**, a fifth lens **5001**, a sixth lens **6001**, and a seventh lens **7001**.

[0153] The first lens **1001** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2001** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3001** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The fourth lens **4001** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5001** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6001** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6001**. The seventh lens **7001** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7001**.

[0154] The optical imaging system **1** further includes a stop, a filter **8001**, and an image sensor **9001**. The stop is disposed between the first lens **1001** and the second lens **2001** to adjust an amount of light incident on the image sensor **9001**. The filter **8001** is disposed between the seventh lens **7001** and the image sensor **9001** to block infrared rays. The image sensor **9001** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **1**, the stop is disposed at a distance of 0.9035 mm from the object-side surface of the first lens **1001** toward the imaging plane of the optical imaging system **1**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 1 listed in Table 57 that appears later in this application.

[0155] Table 1 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **1**, and Table 2 below shows aspherical surface coefficients of the lenses of FIG. **1**.

TABLE-US-00001 TABLE 1 Surface Radius of Thickness/ Index of Abbe Effective Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.0536 0.9035 1.546 56.114 1.568 S2 Lens 8.6990 0.1210 1.513 (Stop) S3 Second 5.7984 0.2300 1.669 20.353 1.411 S4 Lens 3.2822

0.3720 1.251 S5 Third 18.2423 0.5020 1.546 56.114 1.280 S6 Lens −30.8318 0.1292 1.403 S7 Fourth 9.4556 0.2600 1.669 20.353 1.421 S8 Lens 6.8529 0.2842 1.592 S9 Fifth 114.7177 0.3399 1.669 20.353 1.703 S10 Lens 7.7503 0.2357 1.957 S11 Sixth 3.8296 0.8015 1.546 56.114 2.275 S12 Lens −2.3157 0.5095 2.531 S13 Seventh −2.7231 0.3800 1.546 56.114 3.250 S14 Lens 2.7638 0.1236 3.501 S15 Filter Infinity 0.1100 1.519 64.197 3.788 S16 Infinity 0.6860 3.823 S17 Imaging Infinity 0.0120 4.187 Plane

TABLE-US-00002 TABLE 2 K A B C D E F G H J S1 −1.0628 0.0139 0.0094 −0.0141 0.0168 −0.0121 0.0052 −0.0012 0.0001 0 S2 10.994 −0.0496 0.0432 −0.0268 0.0108 −0.0042 0.0015 −0.0004 3E−05 0 S3 0 0 0 0 0 0 0 0 0 0 S4 −1.5785 −0.0696 0.0645 0.0114 −0.0726 0.0789 −0.0411 0.0103 −0.0006 0 S5 0 −0.025 0.0128 −0.0683 0.1144 −0.1136 0.0622 −0.0169 0.0018 0 S6 −95 −0.0612 −0.0021 0.0182 −0.0574 0.0781 −0.0583 0.0229 −0.0037 0 S7 0 −0.1305 0.0429 −0.1213 0.1851 −0.1579 0.0797 −0.0225 0.0028 0 S8 0 −0.1024 0.076 −0.1473 0.1804 −0.1345 0.0601 −0.015 0.0016 0 S9 0 −0.1299 0.161 −0.1553 0.1065 −0.0538 0.0179 −0.0035 0.0003 0 S10 3.6183 −0.1952 0.1484 −0.1106 0.0696 −0.0319 0.0091 −0.0014 9E−05 0 S11 −19.534 −0.0262 −0.0142 0.0017 0.002 −0.0013 0.0003 −3E−05 6E−07 0 S12 −0.7774 0.0934 −0.0701 0.0245 −0.0058 0.0012 0.0002 2E−05 −7E−07 0 S13 −17.906 −0.104 0.0087 0.0102 −0.0036 0.0006 −5E−05 2E−06 −4E−08 0 S14 −0.5975 −0.11 0.0366 −0.009 0.0016 −0.0002 2E−05 −2E−06 6E−08 −1.12E−09

Second Example

[0156] FIG. **3** is a view illustrating a second example of an optical imaging system, and FIG. **4** illustrates aberration curves of the optical imaging system of FIG. **3**.

[0157] An optical imaging system **2** includes a first lens **1002**, a second lens **2002**, a third lens **3002**, a fourth lens **4002**, a fifth lens **5002**, a sixth lens **6002**, and a seventh lens **7002**.

[0158] The first lens **1002** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2002** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens **3002** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4002** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens **5002** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6002** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6002**. The seventh lens **7002** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7002**, and one inflection point is formed on the image-side surface of the seventh lens **7002**.

[0159] The optical imaging system **2** further includes a stop, a filter **8002**, and an image sensor **9002**. The stop is disposed between the second lens **2002** and the third lens **3002** to adjust an amount of light incident on the image sensor **9002**. The filter **8002** is disposed between the seventh lens **7002** and the image sensor **9002** to block infrared rays. The image sensor **9002** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **3**, the stop is disposed at distance of 1.2514 mm from the object-side surface of the first lens **1002** toward the imaging plane of the optical imaging system **2**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 2 listed in Table 57 that appears later in this application.

[0160] Table 3 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **3**, and Table 4 below shows aspherical surface coefficients of the lenses of FIG. **3**.

TABLE-US-00003 TABLE 3 Surface Radius of Thickness/ Index of Abbe Effective Aperture No.

Element Curvature Distance Refraction Number Radius S1 First 2.0977 0.4848 1.546 56.114 1.410 S2 Lens 3.2123 0.1235 1.350 S3 Second 2.8957 0.6231 1.546 56.114 1.310 S4 Lens −16.0261 0.0200 1.271 S5 (Stop) 4.6472 0.2000 1.679 19.236 1.157 S6 Third 2.3076 0.6031 1.095 Lens S7 Fourth −1200.00 0.2984 1.679 19.236 1.270 S8 Lens −1200.00 0.2107 1.456 S9 Fifth 3.3656 0.3072 1.546 56.114 1.712 S10 Lens 3.2933 0.2365 2.000 S11 Sixth 3.2587 0.3776 1.679 19.236 2.150 S12 Lens 2.6817 0.1409 2.500 S13 Seventh 1.5589 0.5411 1.537 53.955 2.871 S14 Lens 1.3718 0.2430 3.050 S15 Filter Infinity 0.1100 1.519 64.166 3.347 S16 Infinity 0.6673 3.379 S17 Imaging Infinity 0.0026 3.708 Plane

TABLE-US-00004 TABLE 4 K A B C D E F G H J S1 −7.583 0.0707 −0.0815 0.0542 −0.0479 0.0209 −0.0011 −0.0013 0.0002 0 S2 −20.327 −0.0052 −0.1116 0.0603 0.0221 −0.0313 0.0098 −0.0002 −0.0003 0 S3 −0.2671 −0.0365 −0.0311 −0.0159 0.08 −0.0164 −0.04 0.0265 −0.0051 0 S4 0 0.0221 −0.096 0.0722 0.0909 −0.2138 0.1659 −0.0596 0.0083 0 S5 −4.5253 −0.0697 0.0432 −0.146 0.4306 −0.6073 0.4481 −0.1664 0.0247 0 S6 0.5431 −0.098 0.1133 −0.1737 0.2753 −0.3038 0.2109 −0.0818 0.0149 0 S7 0 −0.0194 −0.0742 0.1045 −0.1099 0.1045 −0.0888 0.0459 −0.0098 0 S8 0 −0.0129 −0.0975 0.0464 0.0472 −0.0702 0.033 −0.0054 0 0 S9 −43.017 0.1335 −0.1604 0.0703 −0.0277 0.0168 −0.01 0.003 −0.0003 0 S10 −5.2037 −0.0285 0.0684 −0.1295 0.1018 −0.0463 0.0125 −0.0018 0.0001 0 S11 −1.699 0.0274 −0.1873 0.1887 −0.126 0.0512 −0.0118 0.0014 −7E−05 0 S12 −0.0013 −0.0788 −0.0314 0.0355 −0.0206 0.0072 −0.0014 0.0001 −6E−06 0 S13 −0.8015 −0.4138 0.198 −0.0635 0.0157 −0.003 0.0004 −4E−05 2E−06 −5E−08 S14 −1.2781 −0.3 0.1664 −0.0696 0.021 −0.0044 0.0006 −5E−05 3E−06 −5E−08

Third Example

[0161] FIG. **5** is a view illustrating a third example of an optical imaging system, and FIG. **6** illustrates aberration curves of the optical imaging system of FIG. **5**.

[0162] An optical imaging system **3** includes a first lens **1003**, a second lens **2003**, a third lens **3003**, a fourth lens **4003**, a fifth lens **5003**, a sixth lens **6003**, and a seventh lens **7003**.

[0163] The first lens **1003** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2003** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens **3003** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4003** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens **5003** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6003** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6003**. The seventh lens **7003** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7003**, and one inflection point is formed on the image-side surface of the seventh lens **7003**.

[0164] The optical imaging system **3** further includes a stop, a filter **8003**, and an image sensor **9003**. The stop is disposed between the second lens **2003** and the third lens **3003** to adjust an amount of light incident on the image sensor **9003**. The filter **8003** is disposed between the seventh lens **7003** and the image sensor **9003** to block infrared rays. The image sensor **9003** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **5**, the stop is disposed at a distance of 1.4250 mm from the object-side surface of the first lens **1003** toward the imaging plane of the optical imaging system **3**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 3 listed in Table 57 that appears later in this application.

[0165] Table 5 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **5**, and Table 6 below shows aspherical surface coefficients of the lenses of

FIG. **5**.

TABLE-US-00005 TABLE 5 Surface Radius of Thickness/ Index of Abbe Effective Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.3706 0.5431 1.546 56.114 1.572 S2 Lens 3.8377 0.1516 1.517 S3 Second 3.4329 0.7078 1.546 56.114 1.478 S4 Lens −17.0251 0.0225 1.428 S5 (Stop) 5.1429 0.2247 1.679 19.236 1.300 S6 Third 2.5333 0.5888 1.230 Lens S7 Fourth −1446.167 0.3404 1.679 19.236 1.404 S8 Lens −1446.167 0.2070 1.600 S9 Fifth 3.6434 0.3264 1.546 56.114 1.857 S10 Lens 3.8224 0.3171 2.199 S11 Sixth 3.8509 0.4406 1.679 19.236 2.415 S12 Lens 3.0494 0.1774 2.808 S13 Seventh 1.7430 0.6133 1.537 53.955 3.115 S14 Lens 1.5635 0.2466 3.314 S15 Filter Infinity 0.1100 1.519 64.166 3.655 S16 Infinity 0.8047 3.688 S17 Imaging Infinity 0.0051 4.075 Plane

TABLE-US-00006 TABLE 6 K A B C D E F G H J S1 −7.5196 0.0476 −0.039 0.0108 −0.0002 −0.006 0.0045 −0.0012 0.0001 0 S2 −19.661 −0.0106 −0.0481 0.0183 0.0105 −0.0109 0.0039 −0.0006 3E−05 0 S3 0.042 −0.0249 −0.0196 0.0094 0.0041 0.0108 −0.014 0.0056 −0.0008 0 S4 0 0.0098 −0.0507 0.0341 0.0229 −0.0518 0.0341 −0.0103 0.0012 0 S5 −5.6502 −0.0476 0.0152 −0.0398 0.11 −0.1327 0.082 −0.0252 0.0031 0 S6 0.5327 −0.067 0.0583 −0.0705 0.0922 −0.0854 0.0499 −0.0161 0.0024 0 S7 0 −0.0158 −0.0083 −0.0305 0.0756 −0.0736 0.035 −0.0077 0.0005 0 S8 0 −0.0099 −0.0427 0.0077 0.0285 −0.0272 0.01 −0.0013 0 0 S9 −44.395 0.1048 −0.1251 0.08 −0.0437 0.0187 −0.0058 0.001 −7E−05 0 S10 −4.0715 −0.0175 0.0211 −0.0368 0.0252 −0.01 0.0024 −0.0003 2E−05 0 S11 −1.1211 0.0034 −0.0742 0.0637 −0.0381 0.0134 −0.0026 0.0003 −1E−05 0 S12 0.0464 −0.092 0.0339 −0.0168 0.0044 −0.0005 1E−05 3E−06 −2E−07 0 S13 −0.795 −0.2987 0.11 −0.0259 0.0046 −0.0007 7E−05 −5E−06 2E−07 −5E−09 S14 −1.3233 −0.199 0.0846 −0.0285 0.0073 −0.0013 0.0002 −1E−05 5E−07 −9E−09

Fourth Example

[0166] FIG. **7** is a view illustrating a fourth example of an optical imaging system, and FIG. **8** illustrates aberration curves of the optical imaging system of FIG. **7**.

[0167] An optical imaging system **4** includes a first lens **1004**, a second lens **2004**, a third lens **3004**, a fourth lens **4004**, a fifth lens **5004**, a sixth lens **6004**, and a seventh lens **7004**.

[0168] The first lens **1004** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2004** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens **3004** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4004** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens **5004** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6004** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6004**. The seventh lens **7004** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7004**, and one inflection point is formed on the image-side surface of the seventh lens **7004**.

[0169] The optical imaging system **4** further includes a stop, a filter **8004**, and an image sensor **9004**. The stop is disposed between the second lens **2004** and the third lens **3004** to adjust an amount of light incident on the image sensor **9004**. The filter **8004** is disposed between the seventh lens **7004** and the image sensor **9004** to block infrared rays. The image sensor **9004** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **7**, the stop is disposed at a distance of 1.1686 mm from the object-side surface of the first lens **1004** toward the imaging plane of the optical imaging system **4**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 4 listed in Table 57 that appears later in this application.

[0170] Table 7 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **7**, and Table 8 below shows aspherical surface coefficients of the lenses of FIG. **7**.

TABLE-US-00007 TABLE 7

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.9512	0.4488	1.546	56.114	1.307
S2	Lens	3.1152	0.1260			1.253
S3	Second	2.8686	0.5753	1.546	56.114	1.214
S4	Lens	−12.9825	0.0186			1.180
S5 (Stop)		4.5064	0.1856	1.679	19.236	1.074
S6	Third Lens	2.1969	0.5197			1.016
S7	Fourth	−2108.865	0.2796	1.679	19.236	1.179
S8	Lens	−6755.436	0.1715			1.338
S9	Fifth	3.1135	0.2734	1.546	56.114	1.528
S10	Lens	3.2672	0.2417			1.808
S11	Sixth	3.2228	0.3650	1.679	19.236	1.996
S12	Lens	2.5388	0.1438			2.320
S13	Seventh	1.4451	0.5122	1.537	53.955	2.500
S14	Lens	1.2680	0.2501			2.738
S15	Filter	Infinity	0.1100	1.519	64.166	2.940
S16		Infinity	0.5924			2.971
S17	Imaging Plane	Infinity	0.0054			3.251

TABLE-US-00008 TABLE 8

	K	A	B	C	D	E	F	G	H	J
S1	−7.5279	0.0857	−0.105	0.0528	−0.0256	−0.0221	0.0379	−0.0166	0.0023	0
S2	−19.893	−0.0142	−0.1337	0.0682	0.0621	−0.0783	0.0306	−0.0031	−0.0006	0
S3	−0.0142	−0.0449	−0.0418	−0.0147	0.1136	0.012	−0.1333	0.0892	−0.0193	0
S4	0	0.0281	−0.189	0.276	−0.0808	−0.2297	0.2908	−0.1382	0.024	0
S5	−6.2325	−0.0763	−0.0054	−0.0795	0.6054	−1.1875	1.107	−0.5047	0.0912	0
S6	0.4782	−0.115	0.1396	−0.2676	0.5637	−0.7991	0.6898	−0.325	0.0682	0
S7	0	−0.0188	−0.0772	0.0717	0.0184	−0.081	0.0225	0.0277	−0.0139	0
S8	0	−0.0127	−0.1356	0.0837	0.0781	−0.1502	0.0847	−0.0163	0	0
S9	−49.08	0.1815	−0.3205	0.2837	−0.2161	0.1317	−0.0595	0.0158	−0.0017	0
S10	−5.4303	−0.0205	0.025	−0.1003	0.1046	−0.0624	0.0222	−0.0043	0.0003	0
S11	−1.136	0.0314	−0.2615	0.3261	−0.2695	0.133	−0.0369	0.0053	−0.0003	0
S12	0.0272	−0.1293	0.0241	5E−05	−0.0123	0.0085	−0.0024	0.0003	−2E−05	0
S13	−0.8	−0.5247	0.2994	−0.1227	0.0414	−0.0108	0.002	−0.0002	2E−05	−4E−07
S14	−1.3207	−0.3666	0.2425	−0.1248	0.0468	−0.0121	0.002	−0.0002	1E−05	−3E−07

Fifth Example

[0171] FIG. **9** is a view illustrating a fifth example of an optical imaging system, and FIG. **10** illustrates aberration curves of the optical imaging system of FIG. **9**.

[0172] An optical imaging system **5** includes a first lens **1005**, a second lens **2005**, a third lens **3005**, a fourth lens **4005**, a fifth lens **5005**, a sixth lens **6005**, and a seventh lens **7005**.

[0173] The first lens **1005** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2005** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3005** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4005** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5005** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6005** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6005**. The seventh lens **7005** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7005**.

[0174] The optical imaging system **5** further includes a stop, a filter **8005**, and an image sensor **9005**. The stop is disposed between the first lens **1005** and the second lens **2005** to adjust an amount of light incident on the image sensor **9005**. The filter **8005** is disposed between the seventh lens **7005** and the image sensor **9005** to block infrared rays. The image sensor **9005** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **9**, the stop is disposed at distance of 0.3830 mm from the object-side surface of the first lens **1005** toward the imaging plane of the optical imaging system **5**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 5 listed in Table 57 that appears later in this application.

[0175] Table 9 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **9**, and Table 10 below shows aspherical surface coefficients of the lenses of FIG. **9**.

TABLE-US-00009 TABLE 9

Surface No.	Element	Radius of Curvature	Thickness/Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.1824	0.3329	1.546	56.114	1.380
S2	Lens	1.9439	0.0500			1.369
S3 (Stop)		1.6857	0.7322	1.546	56.114	1.335
S4	Second Lens	28.3727	0.0500			1.264
S5	Third	7.1536	0.2200	1.679	19.236	1.185
S6	Lens	2.9223	0.4264			1.050
S7	Fourth	46.9146	0.3121	1.646	23.528	1.112
S8	Lens	17.5860	0.2616			1.268
S9	Fifth	2.2655	0.2700	1.646	23.528	1.774
S10	Lens	2.3143	0.3731			1.839
S11	Sixth	8.5186	0.6078	1.546	56.114	2.160
S12	Lens	−1.9871	0.3782			2.308
S13	Seventh	−4.7165	0.3600	1.546	56.114	2.780
S14	Lens	1.8919	0.1457			2.998
S15	Filter	Infinity	0.1100	1.519	64.166	3.353
S16		Infinity	0.6600			3.385
S17	Imaging Plane	Infinity	0.0100			3.712

TABLE-US-00010 TABLE 10

	K	A	B	C	D	E	F	G	H
S1	−3.5715	0.0005	0.0011	−0.0181	0.0025	0.0107	−0.0084	0.0026	−0.0003
S2	−9.1496	−0.0513	−0.0055	0.0116	0.0161	−0.0207	0.0078	−0.001	0
S3	−2.5622	−0.0879	0.1115	−0.1204	0.1625	−0.1325	0.0578	−0.0118	0.0006
S4	−90	−0.078	0.2103	−0.4384	0.6397	−0.6153	0.3736	−0.1288	0.0189
S5	0	−0.1133	0.2975	−0.5447	0.7496	−0.7199	0.4525	−0.1642	0.0257
S6	4.6946	−0.0705	0.1434	−0.2144	0.1998	−0.0956	−0.0142	0.0399	−0.0137
S7	0	−0.0972	0.1221	−0.3303	0.5457	−0.6222	0.4555	−0.1995	0.0405
S8	0	−0.1596	0.2027	−0.3281	0.3412	−0.2472	0.1212	−0.0385	0.0064
S9	−18.27	−0.0564	−0.0069	0.0518	−0.0566	0.0228	−0.0011	−0.0019	0.0004
S10	−15.127	−0.0603	−0.0145	0.0594	−0.0601	0.0318	−0.0096	0.0015	−1E−04
S11	0	0.0027	−0.0398	0.025	−0.0137	0.005	−0.001	1E−04	−4E−06
S12	−1.1693	0.1224	−0.1006	0.0535	−0.0195	0.005	−0.0008	8E−05	−3E−06
S13	−4.4446	−0.097	−0.0137	0.0358	−0.0141	0.0028	−0.0003	2E−05	−5E−07
S14	−8.7431	−0.0906	0.0342	−0.009	0.0017	−0.0002	2E−05	−1E−06	3E−08

Sixth Example

[0176] FIG. **11** is a view illustrating a sixth example of an optical imaging system, and FIG. **12** illustrates aberration curves of the optical imaging system of FIG. **11**.

[0177] An optical imaging system **6** includes a first lens **1006**, a second lens **2006**, a third lens **3006**, a fourth lens **4006**, a fifth lens **5006**, a sixth lens **6006**, and a seventh lens **7006**.

[0178] The first lens **1006** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2006** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3006** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4006** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5006** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6006** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6006**. The seventh lens **7006** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7006**.

[0179] The optical imaging system **6** further includes a stop, a filter **8006**, and an image sensor **9006**. The stop is disposed between the first lens **1006** and the second lens **2006** to adjust an amount of light incident on the image sensor **9006**. The filter **8006** is disposed between the seventh lens **7006** and the image sensor **9006** to block infrared rays. The image sensor **9006** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **11**, the stop is disposed at a distance of 0.7312 mm from the object-side surface of the first lens **1006** toward the imaging plane of the optical imaging system **6**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 6 listed in Table 57 that appears later in this application.

[0180] Table 11 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **11**, and Table 12 below shows aspherical surface coefficients of the lenses of FIG. **11**.

TABLE-US-00011 TABLE 11 Thick- ness/ Effective Surface Radius of Dis- Index of Abbe Aperture No. Element Curvature tance Refraction Number Radius S1 First 1.73233 0.73124 1.546 56.114 1.250 S2 Lens 12.53699 0.07002 1.181 (Stop) S3 Second 5.58930 0.20000 1.667 20.353 1.147 S4 Lens 2.57397 0.39715 1.100 S5 Third 8.06552 0.38474 1.546 56.114 1.128 S6 Lens 7.83668 0.19259 1.247 S7 Fourth 6.68716 0.24423 1.546 56.114 1.276 S8 Lens 30.32847 0.27130 1.374 S9 Fifth −3.28742 0.24968 1.667 20.353 1.481 S10 Lens −4.51593 0.13884 1.734 S11 Sixth 5.67988 0.51987 1.546 56.114 2.150 S12 Lens −1.89003 0.31663 2.318 S13 Seventh −3.93255 0.30000 1.546 56.114 2.640 S14 Lens 1.74183 0.19371 2.747 S15 Filter Infinity 0.11000 1.518 64.166 3.146 S16 Infinity 0.77000 S17 Imaging Infinity 0.01000 3.536 3.177 Plane

TABLE-US-00012 TABLE 12 K A B C D E F G H J S1 −0.7464 0.0139 0.0344 −0.0749 0.1029 −0.0706 0.0173 0.0042 −0.0023 0 S2 36.669 −0.0823 0.195 −0.3067 0.3634 −0.323 0.1902 −0.0632 0.0086 0 S3 −1.3559 −0.1603 0.3305 −0.4059 0.3324 −0.1787 0.0673 −0.0166 0.0018 0 S4 −0.4109 −0.0907 0.1444 0.1155 −0.7969 1.5009 −1.4406 0.7219 −0.147 0 S5 0 −0.0739 0.0463 −0.1203 0.1165 −0.0578 −0.0089 0.0233 −0.0057 0 S6 0 −0.0932 0.0034 0.0521 −0.1827 0.2457 −0.2173 0.1126 −0.0241 0 S7 25.148 −0.1235 −0.1887 0.3763 −0.554 0.6731 −0.5796 0.2782 −0.0538 0 S8 −99 −9E−05 −0.3274 0.3588 −0.3195 0.3451 −0.2608 0.0995 −0.0144 0 S9 −70.894 0.0205 0.0483 −0.5284 0.7583 −0.4915 0.1636 −0.0271 0.0018 0 S10 2.2832 0.1759 −0.3448 0.2283 −0.0716 0.011 −0.0007 −4E−06 1E−06 0 S11 −99 0.1188 −0.2169 0.1675 −0.0871 0.0276 −0.0049 0.0005 −2E−05 0 S12 −3.3067 0.1644 −0.1849 0.1159 −0.049 0.0138 −0.0024 0.0002 −9E−06 0 S13 −2.4772 −0.1026 −0.0482 0.074 −0.0308 0.0067 −0.0008 6E−05 −2E−06 0 S14 −1.1028 −0.2935 0.2033 −0.1127 0.0457 −0.0129 0.0024 −0.0003 2E−05 −5E−07

Seventh Example

[0181] FIG. **13** is a view illustrating a seventh example of an optical imaging system, and FIG. **14** illustrates aberration curves of the optical imaging system of FIG. **13**.

[0182] An optical imaging system **7** includes a first lens **1007**, a second lens **2007**, a third lens **3007**, a fourth lens **4007**, a fifth lens **5007**, a sixth lens **6007**, and a seventh lens **7007**.

[0183] The first lens **1007** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2007** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens **3007** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4007** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5007** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6007** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6007**. The seventh lens **7007** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7007**, and one inflection point is formed on the image-side surface of the seventh lens **7007**.

[0184] The optical imaging system **7** further includes a stop, a filter **8007**, and an image sensor **9007**. The stop is disposed between the second lens **2007** and the third lens **3007** to adjust an amount of light incident on the image sensor **9007**. The filter **8007** is disposed between the seventh lens **7007** and the image sensor **9007** to block infrared rays. The image sensor **9007** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **13**, the stop is disposed at a distance of 1.1577 mm from the object-side surface of the first lens **1007** toward the imaging plane of the optical imaging system **7**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 7 listed in Table 57 that appears later in this application.

[0185] Table 13 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **13**, and Table 14 below shows aspherical surface coefficients of the lenses of FIG. **13**.

TABLE-US-00013 TABLE 13 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.141 0.481 1.546 56.114 1.450 S2 Lens 3.251 0.110 1.350 S3 Second 3.253 0.542 1.546 56.114 1.285 S4 Lens −15.773 0.025 1.232 S5 (Stop) 8.425 0.230 1.679 19.236 1.157 S6 Third 3.514 0.625 1.095 Lens S7 Fourth 25.986 0.296 1.679 19.236 1.265 S8 Lens 15.894 0.230 1.452 S9 Fifth 3.048 0.400 1.546 56.114 1.675 S10 Lens 3.616 0.290 2.092 S11 Sixth 3.762 0.400 1.679 19.236 2.153 S12 Lens 2.792 0.204 2.476 S13 Seventh 1.614 0.510 1.537 53.955 2.938 S14 Lens 1.326 0.196 3.102 S15 Filter Infinity 0.110 1.518 64.197 3.420 S16 Infinity 0.639 3.450 S17 Imaging Infinity 0.011 3.730 Plane

TABLE-US-00014 TABLE 14 K A B C D E F G H J S1 −8.038 0.0707 −0.0797 0.0334 0.0072 −0.0491 0.0465 −0.0186 0.0032 −0.0002 S2 −20.594 −0.0019 −0.1494 0.2041 −0.2922 0.3755 −0.3085 0.1486 −0.0387 0.0042 S3 −0.0908 −0.0339 −0.0641 0.1368 −0.2821 0.4921 −0.4815 0.2605 −0.0746 0.0088 S4 −0.4822 −0.0436 0.1761 −0.3256 0.1999 0.1916 −0.4291 0.3203 −0.1141 0.0162 S5 −1.1841 −0.1073 0.2544 −0.4683 0.4991 −0.2863 0.0565 0.0325 −0.0229 0.0044 S6 0.8733 −0.0693 0.0357 0.2048 −0.8833 1.7328 −1.9742 1.3464 −0.5106 0.083 S7 −0.4999 −0.0314 0.0135 −0.2894 0.9716 −1.7181 1.7923 −1.1152 0.3837 −0.0563 S8 −1E−06 −0.0273 −0.1177 0.212 −0.2544 0.2157 −0.1264 0.0469 −0.0093 0.0007 S9 −41.843 0.1624 −0.3487 0.4016 −0.3105 0.1396 −0.027 −0.0038 0.0026 −0.0003 S10 −5.1424 0.0397 −0.1364 0.1569 −0.1229 0.0633 −0.0212 0.0044 −0.0005 3E−05 S11 −2.1666 0.0356 −0.1809 0.1985 −0.1438 0.0641 −0.0173 0.0028 −0.0002 9E−06 S12 −0.0207 −0.1043 0.0239 −0.0063 −0.0007 0.0007 −3E−06 −4E−05 7E−06 −4E−07 S13 −0.7948 −0.4128 0.1863 −0.0516 0.0101 −0.0015 0.0002 −1E−05 6E−07 −1E−08 S14 −1.3226 −0.3105 0.1713 −0.0712 0.0213 −0.0043 0.0006 −5E−05 2E−06 −5E−08

Eighth Example

[0186] FIG. **15** is a view illustrating an eighth example of an optical imaging system, and FIG. **16** illustrates aberration curves of the optical imaging system of FIG. **15**.

[0187] An optical imaging system **8** includes a first lens **1008**, a second lens **2008**, a third lens **3008**, a fourth lens **4008**, a fifth lens **5008**, a sixth lens **6008**, and a seventh lens **7008**.

[0188] The first lens **1008** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2008** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3008** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4008** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5008** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6008** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6008**. The seventh lens **7008** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7008**.

[0189] The optical imaging system **8** further includes a stop, a filter **8008**, and an image sensor **9008**. The stop is disposed between the second lens **2008** and the third lens **3008** to adjust an amount of light incident on the image sensor **9008**. The filter **8008** is disposed between the seventh lens **7008** and the image sensor **9008** to block infrared rays. The image sensor **9008** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **15**, the stop is disposed at a distance of 1.0800 mm from the object-side surface of the first lens **1008** toward the imaging plane of the optical imaging system **8**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 8 listed in Table 57 that appears later in this application.

[0190] Table 15 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **15**, and Table 16 below shows aspherical surface coefficients of the lenses of FIG. **15**.

TABLE-US-00015 TABLE 15

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.1367244	0.486053	1.546	56.114	1.360
S2	Lens	2.8897668	0.087572			1.331
S3	Second	3.1970288	0.481363	1.546	56.114	1.301
S4	Lens	109.26781	0.025			1.264
S5 (Stop)	Third	8.2164833	0.333222	1.679	19.236	1.218
S6	Lens	3.6430265	0.428709			1.229
S7	Fourth	5.149363	0.374385	1.546	56.114	1.353
S8	Lens	7.9835412	0.393693			1.392
S9	Fifth	3.8134324	0.4	1.546	56.114	1.576
S10	Lens	4.8504303	0.288816			2.010
S11	Sixth	3.8912859	0.4	1.546	56.114	1.916
S12	Lens	3.0824847	0.229858			2.371
S13	Seventh	1.6011178	0.493312	1.546	56.114	2.526
S14	Lens	1.2142472	0.218016			2.787
S15	Filter	Infinity	0.21	1.518	64.197	3.238
S16		Infinity	0.634932			3.316
S17	Imaging Plane	Infinity	0.015			3.728

TABLE-US-00016 TABLE 16

	K	A	B	C	D	E	F	G	H	J
S1	−1	−0.0136	0.0247	−0.0811	0.0925	−0.0456	−0.0139	0.0275	−0.0118	0.0017
S2	−13.222	0.0184	−0.091	−0.0856	0.3055	−0.3421	0.2293	−0.0976	0.0238	−0.0025
S3	−1.2237	−0.0255	0.013	−0.2994	0.6492	−0.662	0.4089	−0.1567	0.0332	−0.0029
S4	−7.0515	−0.018	0.0942	−0.4684	1.1965	−1.7785	1.5984	−0.8543	0.2492	−0.0305
S5	8.9885	−0.0606	0.147	−0.5476	1.4146	−2.2793	2.2356	−1.2976	0.4106	−0.0546
S6	1.6556	−0.053	0.0664	−0.1724	0.4882	−0.9461	1.1092	−0.7563	0.2786	−0.0429
S7	−4.3409	−0.0524	−0.0067	0.1244	−0.3711	0.5503	−0.4701	0.228	−0.0561	0.0052
S8	5.8589	−0.0866	0.13	−0.4361	0.9157	−1.2163	1.0086	−0.506	0.1405	−0.0165
S9	−43.521	0.0853	−0.1755	0.2257	−0.2234	0.1539	−0.0732	0.0221	−0.0038	0.0003
S10	−3.1047	0.0435	−0.1427	0.1592	−0.1109	0.05	−0.0153	0.0031	−0.0004	2E−05
S11	−16.199	0.1264	−0.2435	0.2571	−0.2182	0.1176	−0.0381	0.0072	−0.0007	3E−05
S12	0.1758	−0.0767	0.0734	−0.0745	0.0337	−0.008	0.0011	−7E−05	2E−06	−1E−08
S13	−0.8173	−0.4272	0.2044	−0.0489	0.002	0.0021	−0.0006	8E−05	−6E−06	2E−07
S14	−1.397	−0.3515	0.2336	−0.1198	0.0447	−0.0113	0.0019	−0.0002	1E−05	−3E−07

Ninth Example

[0191] FIG. **17** is a view illustrating a ninth example of an optical imaging system, and FIG. **18** illustrates aberration curves of the optical imaging system of FIG. **17**.

[0192] An optical imaging system **9** includes a first lens **1009**, a second lens **2009**, a third lens **3009**, a fourth lens **4009**, a fifth lens **5009**, a sixth lens **6009**, and a seventh lens **7009**.

[0193] The first lens **1009** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2009** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3009** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4009** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5009** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6009** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6009**. The seventh lens **7009** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7009**, and one inflection point is formed on the image-side surface of the seventh lens **7009**.

[0194] The optical imaging system **9** further includes a stop, a filter **8009**, and an image sensor **9009**. The stop is disposed between the second lens **2009** and the third lens **3009** to adjust an amount of light incident on the image sensor **9009**. The filter **8009** is disposed between the seventh lens **7009** and the image sensor **9009** to block infrared rays. The image sensor **9009** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **17**, the stop is disposed at a distance of 1.0767 mm from the object-side surface of the first lens **1009** toward the imaging plane of the optical imaging system **9**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 9 listed in Table 57 that appears later in this application.

[0195] Table 17 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **17**, and Table 18 below shows aspherical surface coefficients of the lenses of FIG. **17**.

TABLE-US-00017 TABLE 17

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	2.1183051	0.467301	1.546	56.114	1.360
S2	Lens	2.7465072	0.088291			1.343
S3	Second	2.805316	0.495083	1.546	56.114	1.313
S4	Lens	29.972181	0.026058			1.266
S5	(Stop)	5.6204988	0.273577	1.679	19.236	1.212
S6	Third Lens	2.8589333	0.365293			1.199
S7	Fourth	6.0851103	0.415715	1.546	56.114	1.285
S8	Lens	19.143835	0.530007			1.350
S9	Fifth	5.7830909	0.4	1.679	19.236	1.600
S10	Lens	4.5644102	0.188701			2.100
S11	Sixth	2.807724	0.444625	1.546	56.114	1.903
S12	Lens	3.201154	0.276382			2.470
S13	Seventh	1.6500839	0.458527	1.546	56.114	2.646
S14	Lens	1.1944054	0.21044			2.806
S15	Filter	Infinity	0.21	1.518	64.197	3.241
S16		Infinity	0.643292			3.319
S17	Imaging Plane	Infinity	0.006708			3.729

TABLE-US-00018 TABLE 18

	K	A	B	C	D	E	F	G	H	J
S1	−1	−0.0103	0.0078	−0.0588	0.0925	−0.0904	0.0486	−0.0119	0.0004	0.0002
S2	−13.05	0.0258	−0.1274	0.035	0.0617	−0.0405	0.0003	0.0049	−0.0007	−0.0001
S3	−1.2154	−0.0166	−0.0602	−0.0171	0.0625	0.0481	−0.1007	0.0511	−0.0092	0.0002
S4	−7.0515	−0.047	0.2681	−0.8387	1.4546	−1.5426	1.0264	−0.4201	0.0974	−0.0099
S5	8.8287	−0.0982	0.3106	−0.8268	1.4538	−1.7174	1.3464	−0.6715	0.1944	−0.025
S6	1.7217	−0.0695	0.0939	−0.1196	0.1421	−0.2108	0.2773	−0.2257	0.0997	−0.0182
S7	−1.4309	−0.0448	−0.0056	0.0299	−0.0484	−0.0039	0.0856	−0.1013	0.0511	−0.0095
S8	5.8592	−0.0455	−0.0133	0.0337	−0.0729	0.0922	−0.0766	0.0411	−0.0128	0.0018
S9	−43.521	0.0008	−0.0239	0.0222	−0.0173	0.0051	−0.0002	−0.0003	5E−05	5E−06
S10	−11.855	−0.0163	−0.0578	0.0832	−0.067	0.0334	−0.0109	0.0023	−0.0003	1E−05
S11	−16.199	0.1024	−0.1959	0.1931	−0.1564	0.0797	−0.0243	0.0044	−0.0004	2E−05
S12	0.1668	−0.0913	0.11	−0.1075	0.0537	−0.0157	0.0029	−0.0003	2E−05	−6E−07
S13	−0.8022	−0.4375	0.2118	−0.049	0.0016	0.0021	−0.0006	7E−05	−4E−06	1E−07
S14	−1.407	−0.3709	0.2499	−0.1268	0.0461	−0.0114	0.0018	−0.0002	1E−05	−3E−07

Tenth Example

[0196] FIG. **19** is a view illustrating a tenth example of an optical imaging system, and FIG. **20** illustrates aberration curves of the optical imaging system of FIG. **19**.

[0197] An optical imaging system **10** includes a first lens **1010**, a second lens **2010**, a third lens **3010**, a fourth lens **4010**, a fifth lens **5010**, a sixth lens **6010**, and a seventh lens **7010**.

[0198] The first lens **1010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6010**. The seventh lens **7010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7010**.

[0199] The optical imaging system **10** further includes a stop, a filter **8010**, and an image sensor **9010**. The stop is disposed between the second lens **2010** and the third lens **3010** to adjust an amount of light incident on the image sensor **9010**. The filter **8010** is disposed between the seventh lens **7010** and the image sensor **9010** to block infrared rays. The image sensor **9010** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **19**, the stop is disposed at a distance of 1.1782 mm from the object-side surface of the first lens **1010** toward the imaging plane of the optical imaging system **10**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 10 listed in Table 57 that appears later in this application.

[0200] Table 19 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **19**, and Table 20 below shows aspherical surface coefficients of the lenses of FIG. **19**.

TABLE-US-00019 TABLE 19 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.3673866 0.52496 1.546 56.114 1.449 S2 Lens 3.194732 0.05968 1.420 S3 Second 3.4203396 0.567044 1.546 56.114 1.396 S4 Lens 70.973752 0.026546 1.328 S5 (Stop) 7.7783333 0.350469 1.679 19.236 1.286 S6 Third 3.8162291 0.409426 1.313 Lens S7 Fourth 5.5916886 0.369187 1.546 56.114 1.426 S8 Lens 7.9939167 0.39653 1.495 S9 Fifth 4.1003359 0.426003 1.546 56.114 1.707 S10 Lens 5.3728393 0.303377 1.986 S11 Sixth 4.360929 0.424063 1.679 19.236 1.999 S12 Lens 3.5684544 0.223456 2.401 S13 Seventh 1.7878712 0.541352 1.546 56.114 2.495 S14 Lens 1.3975055 0.200463 2.837 S15 Filter Infinity 0.21 1.518 64.197 3.186 S16 Infinity 0.831011 3.256 S17 Imaging Infinity 0.001244 3.785 Plane

TABLE-US-00020 TABLE 20 K A B C D E F G H J S1 −1 −0.006 −0.0037 −0.012 0.012 0.0136 −0.0351 0.0267 −0.009 0.0011 S2 −13.101 0.0148 −0.0546 −0.0452 0.1269 −0.1111 0.056 −0.0175 0.0031 −0.0002 S3 −1.2472 −0.0205 0.0163 −0.1488 0.2518 −0.2171 0.1218 −0.045 0.0097 −0.0009 S4 −7.0515 −0.0205 0.1049 −0.3805 0.7515 −0.8792 0.6257 −0.2656 0.0617 −0.006 S5 8.9156 −0.0324 −0.0045 0.0638 −0.1003 0.0567 0.0072 −0.0234 0.0103 −0.0015 S6 1.6638 −0.0267 −0.1125 0.5983 −1.3543 1.7261 −1.3193 0.5999 −0.1494 0.0157 S7 −4.619 −0.0378 −0.0049 0.0644 −0.1511 0.1787 −0.1225 0.0479 −0.0096 0.0007 S8 5.6116 −0.0667 0.143 −0.4869 0.9313 −1.0571 0.7241 −0.2938 0.0651 −0.006 S9 −44.124 0.0571 −0.0758 0.0454 −0.0145 0.0009 5E−05 0.0002 −8E−05 9E−06 S10 −4.9813 0.0353 −0.0999 0.1024 −0.0647 0.026 −0.0068 0.0011 −0.0001 4E−06 S11 −15.42 0.099 −0.1649 0.1482 −0.1003 0.0426 −0.0109 0.0016 −0.0001 5E−06 S12 0.1791 −0.0585 0.0531 −0.0451 0.0177 −0.0038 0.0005 −4E−05 2E−06 −3E−08 S13 −0.825 −0.303 0.1166 −0.0228 0.0011 0.0005 −0.0001 1E−05 −7E−07 1E−08 S14 −1.3872 −0.2937 0.1877 −0.0863 0.0269 −0.0055 0.0007 −6E−05 3E−06 −5E−08

Eleventh Example

[0201] FIG. **21** is a view illustrating an eleventh example of an optical imaging system, and FIG. **22** illustrates aberration curves of the optical imaging system of FIG. **21**.

[0202] An optical imaging system **11** includes a first lens **1011**, a second lens **2011**, a third lens

3011, a fourth lens **4011**, a fifth lens **5011**, a sixth lens **6011**, and a seventh lens **7011**.

[0203] The first lens **1011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3011** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6011** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6011**. The seventh lens **7011** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7011**, and one inflection point is formed on the image-side surface of the seventh lens **7011**.

[0204] The optical imaging system **11** further includes a stop, a filter **8011**, and an image sensor **9011**. The stop is disposed between the second lens **2011** and the third lens **3011** to adjust an amount of light incident on the image sensor **9011**. The filter **8011** is disposed between the seventh lens **7011** and the image sensor **9011** to block infrared rays. The image sensor **9011** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **21**, the stop is disposed at a distance of 1.2296 mm from the object-side surface of the first lens **1011** toward the imaging plane of the optical imaging system **11**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 11 listed in Table 57 that appears later in this application.

[0205] Table 21 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **21**, and Table 22 below shows aspherical surface coefficients of the lenses of FIG. **21**.

TABLE-US-00021 TABLE 21 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.0759937 0.461868 1.546 56.114 1.450 S2 Lens 2.4626776 0.142767 1.423 S3 Second 2.5138054 0.6 1.546 56.114 1.392 S4 Lens 29.011046 0.025 1.339 S5 (Stop) 8.6847676 0.23 1.679 19.236 1.295 S6 Third 3.5580278 0.403456 1.273 Lens S7 Fourth 4.7911408 0.352214 1.546 56.114 1.378 S8 Lens 7.0752276 0.349153 1.451 S9 Fifth 4.2812245 0.35 1.546 56.114 1.632 S10 Lens 6.1353451 0.360979 2.012 S11 Sixth 4.4147672 0.43 1.679 19.236 2.013 S12 Lens 3.9218806 0.295711 2.303 S13 Seventh 1.740331 0.438887 1.546 56.114 2.548 S14 Lens 1.2235575 0.199966 2.831 S15 Filter Infinity 0.21 1.518 64.197 3.299 S16 Infinity 0.637453 3.370 S17 Imaging Infinity 0.012547 3.731 Plane

TABLE-US-00022 TABLE 22 K A B C D E F G H J S1 −1 −0.0092 0.003 −0.0414 0.0636 −0.0562 0.026 −0.0049 −0.0002 0.0001 S2 −11.557 0.0601 −0.1844 0.2568 −0.3524 0.3604 −0.23 0.0871 −0.018 0.0016 S3 −0.8307 −0.0024 −0.0852 0.1656 −0.3174 0.3977 −0.2764 0.1063 −0.0212 0.0016 S4 33.131 −0.027 0.1754 −0.4193 0.3931 −0.0382 −0.2294 0.1977 −0.0691 0.0091 S5 14.848 −0.09 0.2473 −0.422 0.2881 0.1413 −0.4099 0.3093 −0.1063 0.0142 S6 2.0645 −0.0757 0.0883 0.0177 −0.3102 0.6013 −0.6108 0.357 −0.1119 0.0146 S7 −10.536 −0.0399 −0.0508 0.2144 −0.4431 0.5288 −0.3825 0.1604 −0.034 0.0025 S8 1.3378 −0.0489 −0.0512 0.1032 −0.1013 0.0149 0.0599 −0.0576 0.0222 −0.0032 S9 −44.096 0.0784 −0.1355 0.1317 −0.0913 0.0374 −0.0091 0.001 4E−05 −1E−05 S10 −6.651 0.049 −0.1189 0.1277 −0.0852 0.0342 −0.0083 0.0012 −1E−04 3E−06 S11 −13.816 0.0584 −0.1268 0.1161 −0.0837 0.0379 −0.0102 0.0016 −0.0001 5E−06 S12 1.0596 −0.0574 0.0273 −0.0248 0.0087 −0.0016 0.0002 −9E−06 2E−07 −5E−10 S13 −0.8717 −0.4042 0.1652 −0.0262 −0.0057 0.0037 −0.0008 9E−05 −6E−06 1E−07 S14 −1.3714 −0.3652 0.2385 −0.1205 0.0439 −0.0107 0.0017 −0.0002 9E−06 −2E−07

Twelfth Example

[0206] FIG. **23** is a view illustrating a twelfth example of an optical imaging system, and FIG. **24** illustrates aberration curves of the optical imaging system of FIG. **23**.

[0207] An optical imaging system **12** includes a first lens **1012**, a second lens **2012**, a third lens **3012**, a fourth lens **4012**, a fifth lens **5012**, a sixth lens **6012**, and a seventh lens **7012**.

[0208] The first lens **1012** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2012** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3012** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4012** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5012** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6012** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6012**. The seventh lens **7012** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7012**, and one inflection point is formed on the image-side surface of the seventh lens **7012**.

[0209] The optical imaging system **12** further includes a stop, a filter **8012**, and an image sensor **9012**. The stop is disposed between the second lens **2012** and the third lens **3012** to adjust an amount of light incident on the image sensor **9012**. The filter **8012** is disposed between the seventh lens **7012** and the image sensor **9012** to block infrared rays. The image sensor **9012** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **23**, the stop is disposed at a distance of 1.1793 mm from the object-side surface of the first lens **1012** toward the imaging plane of the optical imaging system **12**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 12 listed in Table 57 that appears later in this application.

[0210] Table 23 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **23**, and Table 24 below shows aspherical surface coefficients of the lenses of FIG. **23**.

TABLE-US-00023 TABLE 23 Sur- Thick- Effective face Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.3878602 0.35 1.546 56.114 1.470 S2 Lens 2.621324 0.10425 1.439 S3 Second 2.5382264 0.7 1.546 56.114 1.405 S4 Lens 36.207255 0.025 1.322 S5 (Stop) 7.7960711 0.23 1.679 19.236 1.287 S6 Third 3.4912075 0.396134 1.325 Lens S7 Fourth 5.4331241 0.505386 1.546 56.114 1.461 S8 Lens 9.3905062 0.443673 1.563 S9 Fifth 4.6788866 0.484222 1.546 56.114 1.772 S10 Lens 9.5732832 0.475792 2.209 S11 Sixth 7.1894204 0.475353 1.679 19.236 2.238 S12 Lens 3.8397107 0.215445 2.557 S13 Seventh 1.8137383 0.559854 1.546 56.114 3.026 S14 Lens 1.3872365 0.230078 3.262 S15 Filter Infinity 0.11 1.518 64.197 3.724 S16 Infinity 0.635 3.763 S17 Imaging Infinity 0.015 4.160 Plane

TABLE-US-00024 TABLE 24 K A B C D E F G H J S1 −0.9977 −0.0046 −0.0139 −0.024 0.0712 −0.0925 0.0658 −0.0257 0.0051 −0.0004 S2 −10.008 0.0879 −0.2624 0.4378 −0.5581 0.4751 −0.2476 0.0753 −0.0122 0.0008 S3 −0.5121 −0.0082 0.0297 −0.2049 0.402 −0.4509 0.3219 −0.1392 0.0325 −0.0031 S4 42.587 −0.0077 −0.0816 0.5107 −1.169 1.3206 −0.8193 0.2839 −0.0514 0.0038 S5 14.891 −0.0724 0.0409 0.4245 −1.2505 1.5731 −1.0799 0.4229 −0.0892 0.0079 S6 1.8054 −0.0886 0.1906 −0.2777 0.2837 −0.2167 0.1131 −0.0327 0.0033 0.0002 S7 −10.152 0.028 −0.4038 1.2639 −2.2626 2.4845 −1.6998 0.7029 −0.16 0.0153 S8 1.7534 −0.0643 0.1117 −0.3935 0.7438 −0.8306 0.565 −0.2306 0.0519 −0.005 S9 −44.62 0.0891 −0.1333 0.115 −0.0807 0.043 −0.017 0.0044 −0.0006 4E−05 S10 −4.9001 0.0976 −0.137 0.0999 −0.0465 0.0132 −0.0022

0.0002 −1E−05 2E−07 S11 −13.159 0.0954 −0.1535 0.1202 −0.0703 0.0264 −0.006 0.0008 −6E−05 2E−06 S12 0.7792 −0.0211 −0.0288 0.0188 −0.0093 0.003 −0.0006 6E−05 −4E−06 9E−08 S13 −0.8786 −0.3003 0.0918 −0.0015 −0.0073 0.0023 −0.0004 3E−05 −1E−06 3E−08 S14 −1.3086 −0.2504 0.1203 −0.0427 0.0114 −0.0021 0.0003 −2E−05 9E−07 −2E−08

Thirteenth Example

[0211] FIG. **25** is a view illustrating a thirteenth example of an optical imaging system, and FIG. **26** illustrates aberration curves of the optical imaging system of FIG. **25**.

[0212] An optical imaging system **13** includes a first lens **1013**, a second lens **2013**, a third lens **3013**, a fourth lens **4013**, a fifth lens **5013**, a sixth lens **6013**, and a seventh lens **7013**.

[0213] The first lens **1013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3013** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6013** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6013**. The seventh lens **7013** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7013**, and one inflection point is formed on the image-side surface of the seventh lens **7013**.

[0214] The optical imaging system **13** further includes a stop, a filter **8013**, and an image sensor **9013**. The stop is disposed between the second lens **2013** and the third lens **3013** to adjust an amount of light incident on the image sensor **9013**. The filter **8013** is disposed between the seventh lens **7013** and the image sensor **9013** to block infrared rays. The image sensor **9013** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **25**, the stop is disposed at a distance of 1.2052 mm from the object-side surface of the first lens **1013** toward the imaging plane of the optical imaging system **13**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 13 listed in Table 57 that appears later in this application.

[0215] Table 25 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **25**, and Table 26 below shows aspherical surface coefficients of the lenses of FIG. **25**.

TABLE-US-00025 TABLE 25

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First Lens	2.3421801	0.35	1.546	56.114	1.570
S2		2.4608227	0.130273			1.582
S3	Second Lens	2.309069	0.7	1.546	56.114	1.578
S4		38.833557	0.025			1.553
S5 (Stop)	Third Lens	7.6526629	0.25298	1.679	19.236	1.474
S6		3.3272174	0.504716			1.325
S7	Fourth Lens	5.6743754	0.346649	1.679	19.236	1.461
S8		6.5811075	0.40646			1.563
S9	Fifth Lens	4.7925509	0.35	1.546	56.114	1.772
S10		10.114446	0.527822			2.209
S11	Sixth Lens	6.1622692	0.43	1.679	19.236	2.238
S12		3.9299512	0.26246			2.557
S13	Seventh Lens	1.9665801	0.564397	1.546	56.114	3.026
S14		1.490457	0.226843			3.262
S15	Filter	Infinity	0.11	1.518	64.197	3.641
S16		Infinity	0.797349			3.681
S17	Imaging Plane	Infinity	0.015			4.171

TABLE-US-00026 TABLE 26

	K	A	B	C	D	E	F	G	H	J
S1	−1	−0.0146	0.01	−0.0522	0.0643	−0.0468	0.0217	−0.0061	0.0009	−6E−05
S2	−9.9316	0.0529	−0.0805	0.0187	0.0015	0.0183	−0.0206	0.0089	−0.0018	0.0001
S3	−0.3035	−0.0312	0.1094	−0.2703	0.3068	−0.1783	0.0505	−0.0026	−0.0019	0.0003
S4	42.587	−0.14	0.5905	−1.3419	1.845	−1.6056	0.8858	−0.299	0.0561	−0.0045
S5	14.878	−0.1698	0.5798	−1.2212	1.6455	−1.4398	0.807	−0.2778	0.0533	−0.0044
S6	2.4782	−0.0974	0.3302	−0.8477	1.4713	−1.6606	1.1916	−0.5238	0.1288	−0.0136
S7	−10.89	−0.0126								

−0.1352 0.3555 −0.5088 0.426 −0.2043 0.0479 −0.0019 −0.0008 S8 2.4559 −0.0438 −0.0219 −0.0159 0.1351 −0.2288 0.1952 −0.0929 0.0235 −0.0025 S9 −44.519 0.0902 −0.1408 0.1256 −0.0754 0.0253 −0.0032 −0.0008 0.0003 −3E−05 S10 2.7864 0.0708 −0.1198 0.1033 −0.057 0.0193 −0.004 0.0005 −3E−05 9E−07 S11 −17.823 0.0653 −0.1084 0.075 −0.0394 0.0131 −0.0026 0.0003 −2E−05 5E−07 S12 0.5903 −0.0273 −0.0096 0.0039 −0.0025 0.001 −0.0002 2E−05 −1E−06 3E−08 S13 −0.8439 −0.2851 0.1057 −0.0237 0.0034 −0.0003 5E−07 2E−06 −1E−07 4E−09 S14 −1.3672 −0.238 0.119 −0.0473 0.0138 −0.0027 0.0003 −3E−05 1E−06 −2E−08

Fourteenth Example

[0216] FIG. **27** is a view illustrating a fourteenth example of an optical imaging system, and FIG. **28** illustrates aberration curves of the optical imaging system of FIG. **27**.

[0217] An optical imaging system **14** includes a first lens **1014**, a second lens **2014**, a third lens **3014**, a fourth lens **4014**, a fifth lens **5014**, a sixth lens **6014**, and a seventh lens **7014**.

[0218] The first lens **1014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2014** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5014** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6014** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6014**. The seventh lens **7014** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7014**.

[0219] The optical imaging system **14** further includes a stop, a filter **8014**, and an image sensor **9014**. The stop is disposed in front of the first lens **1014** to adjust an amount of light incident on the image sensor **9014**. The filter **8014** is disposed between the seventh lens **7014** and the image sensor **9014** to block infrared rays. The image sensor **9014** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **27**, the stop is disposed at a distance of 0.2500 mm from the object-side surface of the first lens **1014** toward the imaging plane of the optical imaging system **14**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 14 listed in Table 57 that appears later in this application.

[0220] Table 27 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **27**, and Table 28 below shows aspherical surface coefficients of the lenses of FIG. **27**.

TABLE-US-00027 TABLE 27 Thick- Effective Surface Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 (Stop) 1.7211 0.6349 1.544 56.114 1.100 S2 First 11.4571 0.1212 1.071 Lens S3 Second 119.1721 0.2033 1.661 20.353 1.057 S4 Lens 4.4758 0.0843 1.043 S5 Third 4.5258 0.3109 1.544 56.114 1.051 S6 Lens 20.6082 0.2158 1.015 S7 Fourth 13.2152 0.2369 1.544 56.114 1.019 S8 Lens 16.2733 0.2103 1.070 S9 Fifth −6.5732 0.4119 1.651 21.494 1.076 S10 Lens −10.4553 0.3710 1.320 S11 Sixth 3.4779 0.6318 1.544 56.114 1.556 S12 Lens 3.1994 0.2672 2.337 S13 Seventh 2.8804 0.5060 1.544 56.114 2.489 S14 Lens 1.7054 0.1384 2.666 S15 Filter Infinity 0.2100 3.102 S16 Infinity 0.5794 3.177 S17 Imaging Infinity 0.0106 3.529 Plane

TABLE-US-00028 TABLE 28 K A B C D E F G H S1 0.0432 −0.0088 0.0131 −0.0627 0.1199 −0.1345 0.077 −0.018 −0.0004 S2 −26.097 −0.0562 0.051 −0.0514 0.0595 −0.0683 0.0462 −0.0139 −7E−05 S3 −99 −0.1283 0.1953 −0.2779 0.5135 −0.8812 0.9662 −0.5723 0.1395 S4 −16.567 −0.0971 0.1552 −0.3608 0.985 −2.059 2.5647 −1.6683 0.4378 S5 −1.6774 −0.0377 0.065

−0.4515 1.687 −3.5163 4.2391 −2.6607 0.6752 S6 57.913 −0.0559 0.0533 −0.341 1.3373 −2.8539 3.4811 −2.2114 0.5781 S7 −66.305 −0.1749 −0.0635 0.0963 −0.2061 0.5819 −0.9 0.6874 −0.1979 S8 19.549 −0.1228 −0.0686 0.0207 0.1647 −0.2695 0.1725 −0.0616 0.0161 S9 29.709 −0.0709 0.0826 −0.3062 0.6009 −0.6459 0.3344 −0.0761 0 S10 −31.338 −0.1255 0.1076 −0.1494 0.1908 −0.1423 0.0506 −0.0065 0 S11 −46.453 0.0038 −0.1455 0.1534 −0.126 0.0705 −0.0225 0.0029 0 S12 −31.504 0.0093 −0.0326 0.0149 −0.0033 0.0003 −1E−05 −7E−07 0 S13 −0.5233 −0.2947 0.1709 −0.0627 0.0154 −0.0025 0.0003 −1E−05 3E−07 S14 −0.8257 −0.2584 0.1353 −0.0565 0.0166 −0.0032 0.0004 −3E−05 7E−07

Fifteenth Example

[0221] FIG. **29** is a view illustrating a fifteenth example of an optical imaging system, and FIG. **30** illustrates aberration curves of the optical imaging system of FIG. **29**.

[0222] An optical imaging system **15** includes a first lens **1015**, a second lens **2015**, a third lens **3015**, a fourth lens **4015**, a fifth lens **5015**, a sixth lens **6015**, and a seventh lens **7015**.

[0223] The first lens **1015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2015** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The fifth lens **5015** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6015**. The seventh lens **7015** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7015**.

[0224] The optical imaging system **15** further includes a stop, a filter **8015**, and an image sensor **9015**. The stop is disposed between the first lens **1015** and the second lens **2015** to adjust an amount of light incident on the image sensor **9015**. The filter **8015** is disposed between the seventh lens **7015** and the image sensor **9015** to block infrared rays. The image sensor **9015** forms an imaging plane on which an image of a subject is formed.

[0225] Table 29 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **29**, and Table 30 below shows aspherical surface coefficients of the lenses of FIG. **29**.

TABLE-US-00029 TABLE 29 Sur- Thick- Effective face Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.7773 0.6238 1.544 56.114 1.217 S2 Lens 6.4566 0.1000 1.158 (Stop) S3 Second 4.4103 0.2363 1.661 20.353 1.157 S4 Lens 2.6584 0.4138 1.184 S5 Third 6.5879 0.4640 1.544 56.114 1.177 S6 Lens 10.5233 0.1777 1.282 S7 Fourth 13.4749 0.3627 1.544 56.114 1.306 S8 Lens −20.2300 0.2325 1.444 S9 Fifth −3.1831 0.2000 1.661 20.353 1.456 S10 Lens −4.2151 0.1000 1.625 S11 Sixth 6.7646 0.6089 1.544 56.114 2.207 S12 Lens −2.8792 0.4211 2.145 S13 Seventh −6.9958 0.3200 1.544 56.114 2.280 S14 Lens 1.6934 0.1485 3.165 S15 Filter Infinity 0.1100 2.850 S16 Infinity 0.7007 2.888 S17 Imaging Infinity −0.0200 3.276 Plane

TABLE-US-00030 TABLE 30 K A B C D E F G H J S1 −0.5383 0.0108 0.0209 −0.0477 0.0729 −0.06 0.0243 −0.0027 −0.0007 0 S2 5.8135 −0.0459 0.0189 0.0248 −0.0559 0.0486 −0.026 0.0094 −0.0019 0 S3 −10.011 −0.085 0.066 0.02 −0.0808 0.0756 −0.0332 0.0069 −0.0006 0 S4 −0.1875 −0.0544 0.0068 0.26 −0.6655 0.9329 −0.7519 0.3313 −0.061 0 S5 0 −0.0569 0.0063 −0.0275 −0.0046 0.0401 −0.0485 0.0264 −0.0053 0 S6 0 −0.0775 −0.0976 0.271 −0.5329 0.5567 −0.3323 0.1128 −0.0176 0 S7 47.015 −0.0863 −0.1024 0.2298 −0.2721 0.1091 0.0392 −0.0378 0.0065 0 S8

−99 −0.0603 −0.0348 0.057 −0.0468 0.0241 −0.007 0.001 −6E−05 0 S9 −99 −0.2672 0.6153 −0.9745 0.9138 −0.5236 0.1786 −0.0332 0.0026 0 S10 −0.0701 0.0268 −0.0377 −0.0253 0.035 −0.0133 0.0024 −0.0002 7E−06 0 S11 −97.721 0.1556 −0.2109 0.1424 −0.0678 0.02 −0.0033 0.0003 −1E−05 0 S12 −1.5998 0.2298 −0.1811 0.0905 −0.0342 0.0088 −0.0014 0.0001 −4E−06 0 S13 4.8341 −0.1142 −0.0024 0.0306 −0.013 0.0027 −0.0003 2E−05 −5E−07 0 S14 −1.0993 −0.2618 0.1449 −0.0599 0.0171 −0.0032 0.0004 −3E−05 1E−06 −2E−08

Sixteenth Example

[0226] FIG. **31** is a view illustrating a sixteenth example of an optical imaging system, and FIG. **32** illustrates aberration curves of the optical imaging system of FIG. **31**.

[0227] An optical imaging system **16** includes a first lens **1016**, a second lens **2016**, a third lens **3016**, a fourth lens **4016**, a fifth lens **5016**, a sixth lens **6016**, and a seventh lens **7016**.

[0228] The first lens **1016** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2016** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3016** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4016** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5016** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6016** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6016**. The seventh lens **7016** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, no inflection point is formed on the object-side surface of the seventh lens **7016**, and one inflection point is formed on the image-side surface of the seventh lens **7016**.

[0229] The optical imaging system **16** further includes a stop, a filter **8016**, and an image sensor **9016**. The stop is disposed between the first lens **1016** and the second lens **2016** to adjust an amount of light incident on the image sensor **9016**. The filter **8016** is disposed between the seventh lens **7016** and the image sensor **9016** to block infrared rays. The image sensor **9016** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **31**, the stop is disposed at a distance of 0.6409 mm from the object-side surface of the first lens **1016** toward the imaging plane of the optical imaging system **16**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 16 listed in Table 57 that appears later in this application.

[0230] Table 31 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **31**, and Table 32 below shows aspherical surface coefficients of the lenses of FIG. **31**.

TABLE-US-00031 TABLE 31 Thick- Effective Surface Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.7977 0.6409 1.544 56.114 1.270 S2 Lens 3.7422 0.1191 1.211 (Stop) S3 Second 3.0573 0.2200 1.661 20.353 1.190 S4 Lens 2.7951 0.3931 1.130 S5 Third 10.6215 0.4640 1.544 56.114 1.153 S6 Lens 9.0266 0.1000 1.289 S7 Fourth 7.9876 0.3621 1.544 56.114 1.328 S8 Lens 138.7678 0.2334 1.454 S9 Fifth −4.1765 0.2198 1.661 20.353 1.518 S10 Lens −4.1394 0.1000 1.656 S11 Sixth 4.6134 0.6089 1.544 56.114 2.000 S12 Lens −3.5921 0.4726 2.038 S13 Seventh −7.0016 0.3200 1.544 56.114 2.049 S14 Lens 1.6938 0.1107 2.685 S15 Filter Infinity 0.2100 2.942 S16 Infinity 0.5300 3.008 S17 Imaging Infinity 0.0200 3.292 Plane

TABLE-US-00032 TABLE 32 K A B C D E F G H J S1 −0.812 0.0136 0.0311 −0.0769 0.1226 −0.1099 0.0531 −0.0116 0.0005 0 S2 −6.6917 −0.0631 0.0174 0.0714 −0.1648 0.1763 −0.1086 0.0376 −0.0059 0 S3 −14.579 −0.0707 0.0068 0.1319 −0.2129 0.173 −0.0715 0.0127 −0.0005 0 S4 −0.188 −0.0614 −0.0138 0.3338 −0.7392 0.9251 −0.6781 0.276 −0.0477 0 S5 0 −0.0572 0.0435

−0.1733 0.2724 −0.2421 0.0931 −0.0042 −0.0038 0 S6 0 −0.1356 −0.0309 0.2183 −0.5547 0.6931 −0.486 0.1856 −0.0304 0 S7 30.023 −0.2107 0.0007 0.1568 −0.2854 0.2586 −0.1154 0.0236 −0.0019 0 S8 −99 −0.1858 −0.0192 0.2616 −0.4111 0.3392 −0.1538 0.0357 −0.0033 0 S9 −98.995 −0.2935 0.5043 −0.5157 0.2657 −0.0658 0.0056 0.0005 −8E−05 0 S10 −0.0701 −0.0775 0.2223 −0.2703 0.1529 −0.0452 0.0073 −0.0006 2E−05 0 S11 −97.878 0.1479 −0.1956 0.1288 −0.0598 0.0172 −0.0028 0.0002 −8E−06 0 S12 1.4166 0.1234 −0.1416 0.087 −0.0341 0.0088 −0.0014 0.0001 −4E−06 0 S13 9.5503 −0.2864 0.1096 0.0149 −0.0214 0.0064 −0.0009 6E−05 −2E−06 0 S14 −1.2786 −0.3076 0.1777 −0.0626 0.0143 −0.0022 0.0002 −1E−05 5E−07 −7E−09

Seventeenth Example

[0231] FIG. **33** is a view illustrating a seventeenth example of an optical imaging system, and FIG. **34** illustrates aberration curves of the optical imaging system of FIG. **33**.

[0232] An optical imaging system **17** includes a first lens **1017**, a second lens **2017**, a third lens **3017**, a fourth lens **4017**, a fifth lens **5017**, a sixth lens **6017**, and a seventh lens **7017**.

[0233] The first lens **1017** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4017** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The fifth lens **5017** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6017**. The seventh lens **7017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7017**.

[0234] The optical imaging system **17** further includes a stop, a filter **8017**, and an image sensor **9017**. The stop is disposed between the second lens **2017** and the third lens **3017** to adjust an amount of light incident on the image sensor **9017**. The filter **8017** is disposed between the seventh lens **7017** and the image sensor **9017** to block infrared rays. The image sensor **9017** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **33**, the stop is disposed at a distance of 0.6811 mm from the object-side surface of the first lens **1017** toward the imaging plane of the optical imaging system **17**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 17 listed in Table 57 that appears later in this application.

[0235] Table 33 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **33**, and Table 34 below shows aspherical surface coefficients of the lenses of FIG. **33**.

TABLE-US-00033 TABLE 33 Thick- Effective Surface Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.5114 0.4703 1.547 56.114 1.072 S2 Lens 6.0961 0.0200 1.048 S3 Second 1.7572 0.1895 1.660 20.400 0.992 S4 Lens 1.2936 0.3871 0.909 (Stop) S5 Third 3.5767 0.1000 1.660 20.400 0.903 S6 Lens 3.3323 0.2006 0.933 S7 Fourth 8.9505 0.6399 1.547 56.114 1.084 S8 Lens −56.3031 0.3491 1.273 S9 Fifth −8.7735 0.1490 1.650 21.494 1.334 S10 Lens −11.1487 0.0575 1.573 S11 Sixth 4.0153 0.5539 1.650 21.494 1.601 S12 Lens 3.7824 0.2451 1.999 S13 Seventh 1.9199 0.5114 1.537 55.711 2.819 S14 Lens 1.4533 0.1829 2.581 S15 Filter Infinity 0.1100 2.895 S16 Infinity 0.5173 2.935 S17 Imaging Infinity 0.0150 3.262 Plane

TABLE-US-00034 TABLE 34 K A B C D E F G H J S1 −0.0872 0.0085 0.0157 −0.0318 0.0507 −0.0457 0.0214 −0.0042 0 0 S2 25.924 −0.1035 0.4079 −0.9853 1.3538 −1.1103 0.5053 −0.1003 0

0 S3 −2.0252 −0.1329 0.5024 −1.13 1.5305 −1.218 0.5394 −0.1043 0 0 S4 −0.1481 −0.094 0.1762 −0.1747 −0.0182 0.4162 −0.4298 0.1625 0 0 S5 0.829 −0.1421 0.1795 −0.2535 0.4392 −0.4879 0.3327 −0.0954 0 0 S6 6.8952 −0.1777 0.1545 −0.1149 0.1034 −0.051 0.0095 −0.0002 0 0 S7 21.918 −0.0605 0.0485 −0.0459 0.0715 −0.0485 0.0135 −0.0013 0 0 S8 25.736 −0.0682 0.0239 −0.012 0.0083 −0.0027 0.0004 −2E−05 0 0 S9 1.6857 −0.1292 0.2433 −0.407 0.387 −0.2241 0.0741 −0.011 0 0 S10 43.884 −0.107 0.1148 −0.1454 0.095 −0.0303 0.0045 −0.0003 0 0 S11 −52.836 0.0701 −0.2199 0.2058 −0.1343 0.0526 −0.0106 0.0009 0 0 S12 0 −0.0577 −0.027 0.0237 −0.0104 0.0019 0.0002 −0.0001 2E−05 −6E−07 S13 −0.9427 −0.3217 0.0977 −0.0029 −0.0058 0.0017 −0.0002 2E−05 −4E−07 0 S14 −1.0048 −0.2798 0.1282 −0.0461 0.0122 −0.0022 0.0002 −1E−05 4E−07 0

Eighteenth Example

[0236] FIG. **35** is a view illustrating an eighteenth example of an optical imaging system, and FIG. **36** illustrates aberration curves of the optical imaging system of FIG. **35**.

[0237] An optical imaging system **18** includes a first lens **1018**, a second lens **2018**, a third lens **3018**, a fourth lens **4018**, a fifth lens **5018**, a sixth lens **6018**, and a seventh lens **7018**.

[0238] The first lens **1018** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2018** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5018** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6018** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6018**. The seventh lens **7018** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7018**.

[0239] The optical imaging system **18** further includes a stop, a filter **8018**, and an image sensor **9018**. The stop is disposed between the second lens **2018** and the third lens **3018** to adjust an amount of light incident on the image sensor **9018**. The filter **8018** is disposed between the seventh lens **7018** and the image sensor **9018** to block infrared rays. The image sensor **9018** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **35**, the stop is disposed at a distance of 1.0512 mm from the object-side surface of the first lens **1018** toward the imaging plane of the optical imaging system **18**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 18 listed in Table 57 that appears later in this application.

[0240] Table 35 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **35**, and Table 36 below shows aspherical surface coefficients of the lenses of FIG. **35**.

TABLE-US-00035 TABLE 35 Thick- Effective Surface Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.8221 0.5822 1.544 56.114 1.275 S2 Lens 4.8276 0.0564 1.231 S3 Second 4.5461 0.3826 1.544 56.114 1.199 S4 Lens 15.5127 0.0300 1.152 S5 (Stop) 3.9113 0.2000 1.661 20.350 1.086 S6 Third 2.2301 0.3911 1.050 Lens S7 Fourth 14.8039 0.3510 1.544 56.114 1.050 S8 Lens 6.0045 0.0516 1.178 S9 Fifth 4.0426 0.2943 1.639 21.525 1.235 S10 Lens 6.0069 0.3029 1.433 S11 Sixth 50.3009 0.5717 1.544 56.114 1.650 S12 Lens −1.4551 0.3562 2.029 S13 Seventh −3.9227 0.3400 1.544 56.114 2.473 S14 Lens 1.8149 0.1800 2.629 S15 Filter Infinity 0.2100 1.518 64.197 S16 Infinity 0.6200 S17 Imaging Infinity 0.0200 Plane

TABLE-US-00036 TABLE 36 K A B C D E F G H S1 −1.7971 0.02 0.0153 −0.0575 0.0794 −0.0689 0.0296 −0.0048 0 S2 0 −0.0249 −0.1102 0.1727 −0.1632 0.1101 −0.0441 0.0076 0 S3 0 0.0215 −0.1293 0.2068 −0.2278 0.2 −0.1022 0.0204 0 S4 72.117 −0.0714 0.2664 −0.6184 0.7522 −0.5313 0.203 −0.0324 0 S5 −15.337 −0.2046 0.4728 −0.8108 0.9542 −0.6926 0.2852 −0.0496 0 S6 −5.3786 −0.102 0.2031 −0.1151 −0.1096 0.3352 −0.285 0.0916 0 S7 0 −0.0443 −0.0061 −0.1088 0.0952 −0.0067 −0.0694 0.0382 0 S8 0 −0.1919 0.079 0.0071 −0.1552 0.1775 −0.0954 0.0212 0 S9 −54.709 −0.2046 −0.0908 0.3474 −0.3213 0.1526 −0.0388 0.0033 0 S10 0 −0.1486 −0.156 0.3054 −0.2298 0.1087 −0.0342 0.0052 0 S11 0 0.0817 −0.1186 −0.0496 0.1291 −0.0835 0.0241 −0.0026 0 S12 −1.7559 0.2122 −0.171 0.0184 0.0388 −0.0196 0.0037 −0.0003 0 S13 −4.6993 0.0063 −0.2121 0.1837 −0.071 0.0154 −0.0019 0.0001 −4E−06 S14 −1.1263 −0.2142 0.0916 −0.0298 0.0072 −0.0012 0.0001 −9E−06 3E−07

Nineteenth Example

[0241] FIG. **37** is a view illustrating a nineteenth example of an optical imaging system, and FIG. **38** illustrates aberration curves of the optical imaging system of FIG. **37**.

[0242] An optical imaging system **19** includes a first lens **1019**, a second lens **2019**, a third lens **3019**, a fourth lens **4019**, a fifth lens **5019**, a sixth lens **6019**, and a seventh lens **7019**.

[0243] The first lens **1019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2019** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5019** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6019** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6019**. The seventh lens **7019** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens **7019**, and two inflection points are formed on the image-side surface of the seventh lens **7019**.

[0244] The optical imaging system **19** further includes a stop, a filter **8019**, and an image sensor **9019**. The stop is disposed between the first lens **1019** and the second lens **2019** to adjust an amount of light incident on the image sensor **9019**. The filter **8019** is disposed between the seventh lens **7019** and the image sensor **9019** to block infrared rays. The image sensor **9019** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **37**, the stop is disposed at a distance of 0.3740 mm from the object-side surface of the first lens **1019** toward the imaging plane of the optical imaging system **19**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 19 listed in Table 57 that appears later in this application.

[0245] Table 37 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **37**, and Table 38 below shows aspherical surface coefficients of the lenses of FIG. **37**.

TABLE-US-00037 TABLE 37 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.1873 0.3243 1.546 56.114 1.450 S2 Lens 1.8391 0.0497 1.441 S3 (Stop) 1.6361 0.7740 1.546 56.114 1.415 S4 Second 30.6063 0.0300 1.354 Lens S5 Third 7.2628 0.2100 1.678 19.236 1.270 S6 Lens 2.9652 0.4149 1.120 S7 Fourth 14.3312 0.3269 1.645 23.528 1.182 S8 Lens 12.1292 0.2502 1.337 S9 Fifth 2.1804 0.2500 1.645 23.528 1.580 S10 Lens 2.1733 0.3831 1.892 S11 Sixth 8.6678 0.6610 1.546 56.114 2.429 S12 Lens −1.9375 0.3110 2.544 S13 Seventh −7.6533 0.3650 1.546 56.114 2.916 S14 Lens 1.6261 0.2200 3.075 S15 Filter Infinity 0.1100 1.518 64.166 3.378 S16 Infinity 0.6351 3.414

S17 Imaging Infinity 0.0049 3.763 Plane
TABLE-US-00038 TABLE 38 K A B C D E F G H S1 −3.7488 0.0012 −0.0066 −0.0004 −0.0198 0.0252 −0.0132 0.0034 −0.0004 S2 −7.1577 −0.061 −0.0104 0.0163 0.0115 −0.0163 0.0063 −0.0009 0 S3 −2.6408 −0.0742 0.0698 −0.0582 0.0727 −0.0412 0.0034 0.0048 −0.0013 S4 −99 −0.0752 0.197 −0.3925 0.5174 −0.4377 0.2286 −0.0663 0.008 S5 0 −0.1076 0.2644 −0.4642 0.6109 −0.5485 0.3128 −0.0997 0.0134 S6 4.364 −0.0584 0.0882 −0.068 −0.0405 0.1629 −0.1817 0.0962 −0.0201 S7 0 −0.0603 0.0743 −0.2389 0.4197 −0.4882 0.353 −0.1472 0.0274 S8 0 −0.1174 0.165 −0.2983 0.348 −0.2864 0.1556 −0.0507 0.0077 S9 −15.429 −0.0562 0.0005 0.0397 −0.0576 0.0355 −0.0117 0.0015 3E−05 S10 −9.1654 −0.1003 0.0623 −0.0379 0.0141 −0.0032 5E−05 0.0002 −3E−05 S11 0 −0.001 −0.0216 0.0157 −0.0111 0.0043 −0.0009 8E−05 −3E−06 S12 −1.7327 0.1074 −0.0935 0.0649 −0.0289 0.0078 −0.0012 0.0001 −4E−06 S13 0.6082 −0.1509 0.0462 0.0036 −0.0043 0.001 −0.0001 6E−06 −2E−07 S14 −8.5925 −0.0951 0.041 −0.0124 0.0026 −0.0004 4E−05 −2E−06 4E−08
Twentieth Example
[0246] FIG. **39** is a view illustrating a twentieth example of an optical imaging system, and FIG. **40** illustrates aberration curves of the optical imaging system of FIG. **39**.
[0247] An optical imaging system **20** includes a first lens **1020**, a second lens **2020**, a third lens **3020**, a fourth lens **4020**, a fifth lens **5020**, a sixth lens **6020**, and a seventh lens **7020**.
[0248] The first lens **1020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2020** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4020** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5020** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6020**. The seventh lens **7020** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens **7020**, and two inflection points are formed on the image-side surface of the seventh lens **7020**.
[0249] The optical imaging system **20** further includes a stop, a filter **8020**, and an image sensor **9020**. The stop is disposed between the first lens **1020** and the second lens **2020** to adjust an amount of light incident on the image sensor **9020**. The filter **8020** is disposed between the seventh lens **7020** and the image sensor **9020** to block infrared rays. The image sensor **9020** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **39**, the stop is disposed at a distance of 1.0601 mm from the object-side surface of the first lens **1020** toward the imaging plane of the optical imaging system **20**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 20 listed in Table 57 that appears later in this application.
[0250] Table 39 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **39**, and Table 40 below shows aspherical surface coefficients of the lenses of FIG. **39**.
TABLE-US-00039 TABLE 39 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.0122 1.0601 1.546 56.114 1.510 S2 Lens 5.8868 0.0711 1.375 (Stop) S3 Second 5.4922 0.2000 1.677 19.238 1.353 S4 Lens 3.7029 0.2969 1.240 S5 Third 9.9760 0.3703 1.546 56.114 1.270 S6 Lens 29.4320 0.1826 1.356 S7 Fourth 8.6841 0.2108 1.667 20.377 1.355 S8 Lens 6.0112 0.2603 1.479 S9 Fifth 5.7445 0.2195 1.619 25.960 1.750 S10 Lens 5.0434 0.3482 1.736 S11 Sixth 4.8476 0.9335 1.546

56.114 2.400 S12 Lens −1.7967 0.3268 2.438 S13 Seventh −1.9512 0.3000 1.546 56.114 2.692 S14 Lens 2.8062 0.1700 3.158 S15 Filter Infinity 0.2100 1.518 64.197 3.744 S16 Infinity 0.6441 3.826 S17 Imaging Infinity −0.0041 4.252 Plane

TABLE-US-00040 TABLE 40 K A B C D E F G H J S1 −1.1332 0.0154 0.0091 −0.0108 0.01 −0.0053 0.0015 −0.0002 −1E−05 0 S2 12.901 −0.0771 0.0315 0.036 −0.0911 0.0858 −0.0454 0.0132 −0.0016 0 S3 9.6379 −0.1255 0.0603 0.0962 −0.2261 0.222 −0.1214 0.036 −0.0045 0 S4 −0.8719 −0.061 0.0524 0.0027 0.0041 −0.069 0.0931 −0.0498 0.0101 0 S5 0 −0.0121 0.0151 −0.088 0.1573 −0.1697 0.1016 −0.0315 0.004 0 S6 −99 −0.0237 −0.0073 0.0365 −0.1182 0.1732 −0.1421 0.0603 −0.0104 0 S7 0 −0.104 −0.0751 0.2741 −0.5665 0.6635 −0.4435 0.1557 −0.0222 0 S8 0 −0.1088 0.0332 −0.0258 −0.0057 0.0249 −0.0175 0.0048 −0.0003 0 S9 0 −0.1824 0.192 −0.1693 0.1022 −0.0423 0.0117 −0.002 0.0002 0 S10 −96.971 −0.1318 0.0661 −0.0223 −0.0048 0.0087 −0.0034 0.0006 −4E−05 0 S11 −34.065 −0.0075 −0.0001 −0.0106 0.0081 −0.0035 0.0008 −1E−04 4E−06 0 S12 −2.7443 0.1131 −0.0726 0.0291 −0.0078 0.0012 −9E−05 1E−06 1E−07 0 S13 −10.221 −0.0393 −0.0198 0.0148 −0.0036 0.0005 −3E−05 1E−06 −2E−08 0 S14 −1.2844 −0.0938 0.0319 −0.0092 0.002 −0.0003 4E−05 −3E−06 1E−07 −3E−09

Twenty-First Example

[0251] FIG. **41** is a view illustrating a twenty-first example of an optical imaging system, and FIG. **42** illustrates aberration curves of the optical imaging system of FIG. **41**.

[0252] An optical imaging system **21** includes a first lens **1021**, a second lens **2021**, a third lens **3021**, a fourth lens **4021**, a fifth lens **5021**, a sixth lens **6021**, and a seventh lens **7021**.

[0253] The first lens **1021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6021**. The seventh lens **7021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7021**, and one inflection point is formed on the image-side surface of the seventh lens **7021**.

[0254] The optical imaging system **21** further includes a stop, a filter **8021**, and an image sensor **9021**. The stop is disposed between the second lens **2021** and the third lens **3021** to adjust an amount of light incident on the image sensor **9021**. The filter **8021** is disposed between the seventh lens **7021** and the image sensor **9021** to block infrared rays. The image sensor **9021** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **41**, the stop is disposed at a distance of 1.1280 mm from the object-side surface of the first lens **1021** toward the imaging plane of the optical imaging system **21**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 21 listed in Table 57 that appears later in this application.

[0255] Table 41 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **41**, and Table 42 below shows aspherical surface coefficients of the lenses of FIG. **41**.

TABLE-US-00041 TABLE 41 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.1378 0.4606 1.546 56.114 1.360 S2 Lens 2.7210 0.0424 1.346 S3 Second 2.7717 0.6000 1.546 56.114 1.322 S4 Lens 33.8379 0.0250 1.253 S5 (Stop) 5.9058 0.2300 1.679 19.236 1.199 S6 Third 2.9580 0.3150

1.193 Lens S7 Fourth 6.7061 0.5156 1.546 56.114 1.246 S8 Lens 15.6197 0.4883 1.350 S9 Fifth 9.4476 0.3912 1.679 19.236 1.600 S10 Lens 5.2667 0.1323 2.100 S11 Sixth 2.4900 0.4534 1.546 56.114 1.951 S12 Lens 2.6058 0.1501 2.440 S13 Seventh 1.4290 0.5074 1.546 56.114 2.691 S14 Lens 1.2861 0.4042 2.841 S15 Filter Infinity 0.2100 1.518 64.197 3.245 S16 Infinity 0.6767 3.316 S17 Imaging Infinity 0.0150 3.733 Plane

TABLE-US-00042 TABLE 42 K A B C D E F G H J S1 −0.9855 −0.0214 0.0439 −0.0925 0.0633 0.0064 −0.0479 0.0372 −0.0126 0.0016 S2 −12.849 0.0234 −0.0441 −0.1546 −0.0352 0.7096 −1.0004 0.6322 −0.1959 0.0242 S3 −1.1002 −0.0276 0.0854 −0.4269 0.4011 0.3152 −0.8128 0.5995 −0.2021 0.0266 S4 −7.367 −0.1684 1.4677 −5.7804 12.64 −16.742 13.734 −6.8183 1.8769 −0.22 S5 9.3187 −0.2245 1.5162 −5.8569 13.059 −17.823 15.121 −7.7778 2.2231 −0.2714 S6 1.6265 −0.0856 0.2704 −0.9806 2.415 −3.7649 3.6777 −2.1905 0.7327 −0.1058 S7 −4.7815 0.0264 −0.5178 1.9131 −4.2532 5.8667 −5.0521 2.6239 −0.7455 0.0886 S8 5.8592 −0.0338 −0.0317 0.0097 0.0291 −0.0644 0.0612 −0.0311 0.0084 −0.0008 S9 −43.521 −0.002 −0.0021 0.0436 −0.1236 0.1389 −0.0871 0.0311 −0.0059 0.0005 S10 −12.729 −0.0608 0.0286 0.0052 −0.0244 0.0182 −0.0074 0.0018 −0.0002 1E−05 S11 −16.199 0.1227 −0.2762 0.2845 −0.2154 0.1043 −0.0311 0.0056 −0.0006 2E−05 S12 0.0242 −0.0902 0.058 −0.0568 0.029 −0.0088 0.0017 −0.0002 2E−05 −5E−07 S13 −0.8394 −0.4114 0.2062 −0.0647 0.0137 −0.0021 0.0003 −2E−05 2E−06 −5E−08 S14 −1.3743 −0.2983 0.1734 −0.0777 0.0258 −0.006 0.0009 −9E−05 5E−06 −1E−07

Twenty-Second Example

[0256] FIG. **43** is a view illustrating a twenty-second example of an optical imaging system, and FIG. **44** illustrates aberration curves of the optical imaging system of FIG. **43**.

[0257] An optical imaging system **22** includes a first lens **1022**, a second lens **2022**, a third lens **3022**, a fourth lens **4022**, a fifth lens **5022**, a sixth lens **6022**, and a seventh lens **7022**.

[0258] The first lens **1022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens **3022** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5022** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6022**. The seventh lens **7022** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7022**.

[0259] The optical imaging system **22** further includes a stop, a filter **8022**, and an image sensor **9022**. The stop is disposed between the second lens **2022** and the third lens **3022** to adjust an amount of light incident on the image sensor **9022**. The filter **8022** is disposed between the seventh lens **7022** and the image sensor **9022** to block infrared rays. The image sensor **9022** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **43**, the stop is disposed at a distance of 1.2251 mm from the object-side surface of the first lens **1022** toward the imaging plane of the optical imaging system **22**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 22 listed in Table 57 that appears later in this application.

[0260] Table 43 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **43**, and Table 44 below shows aspherical surface coefficients of the lenses of FIG. **43**.

TABLE-US-00043 TABLE 43 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.2670 0.4000 1.546 56.114 1.380 S2 Lens 2.6691 0.0251 1.337 S3 Second 2.6238 0.8000 1.546 56.114 1.337 S4 Lens −11.8758 0.0250 1.298 (Stop) S5 Third 19.0032 0.2526 1.679 19.236 1.202 S6 Lens 4.0676 0.4108 1.242 S7 Fourth 6.6991 0.3500 1.679 19.236 1.265 S8 Lens 7.2200 0.4296 1.369 S9 Fifth 21.5310 0.3500 1.546 56.114 1.600 S10 Lens 7.8918 0.0250 1.853 S11 Sixth 2.5030 0.4300 1.546 56.114 1.908 S12 Lens 2.4093 0.2399 2.372 S13 Seventh 1.3275 0.5254 1.546 56.114 2.507 S14 Lens 1.1947 0.3215 2.738 S15 Filter Infinity 0.2100 1.518 64.197 3.205 S16 Infinity 0.6902 3.285 S17 Imaging Infinity 0.0150 3.781 Plane

TABLE-US-00044 TABLE 44 K A B C D E F G H J S1 −1 0.0012 −0.0631 0.0884 −0.0759 −0.0183 0.0824 −0.0602 0.019 −0.0023 S2 −11.056 0.038 0.1354 −0.9871 1.5641 −1.0995 0.3265 0.0108 −0.0287 0.0048 S3 −0.7436 −0.0807 0.579 −1.9725 2.9346 −2.2115 0.8223 −0.0859 −0.0317 0.0076 S4 −7.2488 −0.1245 0.5877 −1.6544 2.8953 −3.2415 2.3118 −1.0127 0.2473 −0.0257 S5 12.337 −0.1601 0.4751 −0.9119 1.1418 −0.9059 0.4213 −0.0904 −0.0021 0.0031 S6 −0.7614 −0.0925 0.1749 −0.2349 0.2283 −0.1666 0.0772 −0.0124 −0.0054 0.0021 S7 −12.018 0.047 −0.734 2.692 −5.8982 8.0627 −6.9318 3.6331 −1.0592 0.1316 S8 5.8397 −0.055 0.0238 −0.178 0.4679 −0.6732 0.5802 −0.2983 0.0845 −0.0101 S9 −43.467 0.0208 −0.0298 0.0644 −0.1157 0.1018 −0.0488 0.0125 −0.0016 7E−05 S10 −10.152 −0.0392 0.0146 0.0256 −0.0749 0.0675 −0.0305 0.0074 −0.0009 5E−05 S11 −16.19 0.0915 −0.1594 0.2045 −0.2282 0.1461 −0.0541 0.0117 −0.0014 7E−05 S12 −0.0871 −0.1985 0.3277 −0.3309 0.1866 −0.0647 0.0141 −0.0019 0.0001 −4E−06 S13 −0.8728 −0.462 0.2873 −0.1331 0.0447 −0.0104 0.0016 −0.0002 9E−06 −2E−07 S14 −1.5388 −0.3036 0.1844 −0.0807 0.0232 −0.0042 0.0005 −3E−05 1E−06 −1E−08

Twenty-Third Example

[0261] FIG. **45** is a view illustrating a twenty-third example of an optical imaging system, and FIG. **46** illustrates aberration curves of the optical imaging system of FIG. **45**.

[0262] An optical imaging system **23** includes a first lens **1023**, a second lens **2023**, a third lens **3023**, a fourth lens **4023**, a fifth lens **5023**, a sixth lens **6023**, and a seventh lens **7023**.

[0263] The first lens **1023** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2023** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3023** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4023** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5023** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6023** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6023**. The seventh lens **7023** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7023**, and one inflection point is formed on the image-side surface of the seventh lens **7023**.

[0264] The optical imaging system **23** further includes a stop, a filter **8023**, and an image sensor **9023**. The stop is disposed between the second lens **2023** and the third lens **3023** to adjust an amount of light incident on the image sensor **9023**. The filter **8023** is disposed between the seventh lens **7023** and the image sensor **9023** to block infrared rays. The image sensor **9023** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **45**, the stop is disposed at a distance of 0.9515 mm from the object-side surface of the first lens **1023** toward the imaging plane of the optical imaging system **23**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 23 listed in Table 57 that appears later in this application.

[0265] Table 45 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **45**, and Table 46 below shows aspherical surface coefficients of the lenses of FIG. **45**.

TABLE-US-00045 TABLE 45 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.7473 0.6964 1.546 56.114 1.280 S2 Lens 9.4084 0.0250 1.247 S3 Second 2.9766 0.2300 1.667 20.353 1.150 S4 Lens 1.9564 0.3428 1.007 (Stop) S5 Third 16.8676 0.2300 1.667 20.353 1.032 S6 Lens 16.0126 0.0294 1.089 S7 Fourth 7.3144 0.3570 1.546 56.114 1.130 S8 Lens 17.3919 0.3708 1.228 S9 Fifth 11.5617 0.3608 1.656 21.525 1.317 S10 Lens 6.9184 0.2917 1.657 S11 Sixth −97.1635 0.5908 1.656 21.525 1.878 S12 Lens 17.2767 0.0301 2.338 S13 Seventh 1.9322 0.8258 1.546 56.114 2.961 S14 Lens 1.7390 0.2207 3.015 S15 Filter Infinity 0.2100 1.518 64.197 3.305 S16 Infinity 0.6356 3.375 S17 Imaging Infinity 0.0143 3.731 Plane

TABLE-US-00046 TABLE 46 K A B C D E F G H J S1 −0.3029 0.0003 0.0248 −0.0645 0.0887 −0.0757 0.0373 −0.0109 0.0014 0 S2 0.9997 −0.0385 0.0595 −0.0639 0.0052 0.0552 −0.0624 0.0296 −0.0054 0 S3 −1.759 −0.0559 0.0769 −0.084 0.0959 −0.0711 0.0309 −0.0026 −0.0012 0 S4 −0.2233 −0.022 −0.0153 0.1358 −0.2648 0.3311 −0.2167 0.051 0.0098 0 S5 −0.8179 −0.0092 −0.0103 −0.1607 0.6303 −1.1881 1.2746 −0.7449 0.1847 0 S6 −0.0005 0.02 −0.1312 0.1142 −0.0014 0.0632 −0.1761 0.1336 −0.0335 0 S7 −31.717 0.0266 −0.0935 −0.0104 0.2126 −0.2049 0.0541 0.02 −0.0098 0 S8 −1.0151 −0.0315 0.0288 −0.0714 0.0935 −0.1394 0.1768 −0.1344 0.0524 −0.0076 S9 0.382 −0.1094 0.0327 −0.0826 0.2138 −0.3162 0.2427 −0.0962 0.0156 0 S10 −27.524 −0.0394 −0.117 0.1628 −0.1238 0.0551 −0.0144 0.0023 −0.0002 0 S11 23.203 0.1802 −0.2793 0.2208 −0.1258 0.0475 −0.0113 0.0016 −0.0001 0 S12 −49.948 0.0336 −0.0362 0.0098 −0.0011 −0.0001 8E−05 −1E−05 6E−07 0 S13 −1.8504 −0.2437 0.1076 −0.031 0.0066 −0.001 0.0001 −6E−06 1E−07 0 S14 −0.8299 −0.173 0.0629 −0.0196 0.0044 −0.0006 6E−05 −3E−06 6E−08 0

Twenty-Fourth Example

[0266] FIG. **47** is a view illustrating a twenty-fourth example of an optical imaging system, and FIG. **48** illustrates aberration curves of the optical imaging system of FIG. **47**.

[0267] An optical imaging system **24** includes a first lens **1024**, a second lens **2024**, a third lens **3024**, a fourth lens **4024**, a fifth lens **5024**, a sixth lens **6024**, and a seventh lens **7024**.

[0268] The first lens **1024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2024** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4024** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5024** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6024** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6024**. The seventh lens **7024** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, no inflection point is formed on the object-side surface of the seventh lens **7024**, and one inflection point is formed on the image-side surface of the seventh lens **7024**.

[0269] The optical imaging system **24** further includes a stop, a filter **8024**, and an image sensor **9024**. The stop is disposed between the first lens **1024** and the second lens **2024** to adjust an amount of light incident on the image sensor **9024**. The filter **8024** is disposed between the seventh lens **7024** and the image sensor **9024** to block infrared rays. The image sensor **9024** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **47**, the stop is disposed at a distance of 0.8574 mm from the object-side surface of the first lens **1024** toward the imaging plane of the optical imaging system **24**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 24 listed in Table 57 that appears later in this application.

[0270] Table 47 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **47**, and Table 48 below shows aspherical surface coefficients of the lenses of FIG. **47**.

TABLE-US-00047 TABLE 47 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.8263 0.7034 1.546 56.114 1.290 S2 Lens 7.7056 0.1540 1.215 S3 (Stop) 4.6213 0.2200 1.679 19.236 1.127 S4 Second 2.7291 0.3146 1.109 Lens S5 Third 6.6824 0.4391 1.546 56.114 1.151 S6 Lens 11.7185 0.1811 1.250 S7 Fourth 6.8604 0.2500 1.679 19.236 1.259 S8 Lens 7.4620 0.4065 1.408 S9 Fifth −9.8497 0.5939 1.546 56.114 1.604 S10 Lens −1.8870 0.0250 1.970 S11 Sixth −41.8807 0.3701 1.546 56.114 2.299 S12 Lens −3.7454 0.2569 2.568 S13 Seventh −2.0634 0.3200 1.546 56.114 2.855 S14 Lens 2.6116 0.1554 3.055 S15 Filter Infinity 0.2100 1.518 64.197 3.346 S16 Infinity 0.6400 3.410 S17 Imaging Infinity 0.0100 3.730 Plane

TABLE-US-00048 TABLE 48 K A B C D E F G H J S1 −1.0945 0.0136 0.0506 −0.1839 0.416 −0.5839 0.51 −0.2705 0.0795 −0.01 S2 3.251 −0.0482 0.0508 −0.085 0.2198 −0.436 0.5133 −0.3477 0.1258 −0.0189 S3 −13.699 −0.1155 0.1942 −0.4376 1.335 −2.7707 3.4839 −2.5718 1.0289 −0.1723 S4 −4.0179 −0.0945 0.2406 −0.7546 2.4023 −4.9111 6.1463 −4.5679 1.8552 −0.3168 S5 −6.6783 −0.0675 0.1229 −0.5308 1.3347 −2.1668 2.2329 −1.4059 0.4923 −0.0729 S6 2.6687 −0.1089 0.0811 −0.1248 0.0166 0.1977 −0.3307 0.2573 −0.1017 0.0162 S7 7.0258 −0.2027 0.0564 −0.0521 0.0446 −0.0418 0.0403 −0.0212 0.001 0.0016 S8 −10.8 −0.1484 0.0297 −0.0692 0.1666 −0.2292 0.2033 −0.1109 0.0326 −0.0038 S9 −26.465 0.0072 −0.0015 −0.1473 0.2748 −0.3047 0.2171 −0.0939 0.0219 −0.0021 S10 −1.4915 0.1141 −0.2124 0.1883 −0.1127 0.0475 −0.0129 0.0021 −0.0002 6E−06 S11 −6.8308 0.0507 −0.1087 0.0643 −0.0416 0.0215 −0.0064 0.0011 −9E−05 3E−06 S12 −10.262 0.0544 0.062 −0.1082 0.0705 −0.0254 0.0054 −0.0007 4E−05 −1E−06 S13 −6.0066 0.0037 −0.0456 0.0731 −0.0405 0.0115 −0.0019 0.0002 −9E−06 2E−07 S14 −0.8095 −0.1128 0.0401 −0.0105 0.0011 0.0002 −7E−05 8E−06 −5E−07 1E−08

Twenty-Fifth Example

[0271] FIG. **49** is a view illustrating a twenty-fifth example of an optical imaging system, and FIG. **50** illustrates aberration curves of the optical imaging system of FIG. **49**.

[0272] An optical imaging system **25** includes a first lens **1025**, a second lens **2025**, a third lens **3025**, a fourth lens **4025**, a fifth lens **5025**, a sixth lens **6025**, and a seventh lens **7025**.

[0273] The first lens **1025** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2025** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3025** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens **4025** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5025** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6025** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6025**. The seventh lens **7025** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, no inflection point is formed on the object-side surface of the seventh lens **7025**, and one inflection point is formed on the image-side surface of the seventh lens **7025**.

[0274] The optical imaging system **25** further includes a stop, a filter **8025**, and an image sensor

9025. The stop is disposed between the second lens **2025** and the third lens **3025** to adjust an amount of light incident on the image sensor **9025**. The filter **8025** is disposed between the seventh lens **7025** and the image sensor **9025** to block infrared rays. The image sensor **9025** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **49**, the stop is disposed at a distance of 0.8722 mm from the object-side surface of the first lens **1025** toward the imaging plane of the optical imaging system **25**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 25 listed in Table 57 that appears later in this application.

[0275] Table 49 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **49**, and Table 50 below shows aspherical surface coefficients of the lenses of FIG. **49**.

TABLE-US-00049 TABLE 49

Surface No.	Element	Radius of Curvature	Thickness/ Distance	Index of Refraction	Abbe Number	Effective Aperture Radius
S1	First	1.7603	0.6172	1.546	56.114	1.100
S2	Lens	14.1233	0.0250			1.040
S3	Second	5.8341	0.2300	1.667	20.353	1.011
S4	Lens (Stop)	3.1227	0.3733			0.919
S5	Third	−49.9417	0.3799	1.546	56.114	0.995
S6	Lens	−15.1870	0.1809			1.096
S7	Fourth	23.3680	0.3032	1.667	20.353	1.124
S8	Lens	12.2098	0.3354			1.309
S9	Fifth	−4.3948	0.4729	1.546	56.114	1.471
S10	Lens	−1.5983	0.0250			1.698
S11	Sixth	−6.0815	0.5447	1.546	56.114	1.822
S12	Lens	−3.0145	0.2724			2.192
S13	Seventh	−6.1494	0.4224	1.546	56.114	2.462
S14	Lens	1.6367	0.1933			2.880
S15	Filter	Infinity	0.2100	1.518	64.197	3.223
S16		Infinity	0.6445			3.300
S17	Imaging Plane	Infinity	0.0099			3.728

TABLE-US-00050 TABLE 50

	K	A	B	C	D	E	F	G	H	J
S1	−1.0054	0.0225	0.0222	−0.0696	0.1604	−0.2238	0.1806	−0.0791	0.0141	0
S2	−1.5097	−0.1275	0.3975	−0.6982	0.6801	−0.322	0.0288	0.029	−0.0076	0
S3	6.0294	−0.163	0.4504	−0.8514	1.0525	−0.8203	0.4235	−0.138	0.0213	0
S4	−0.8846	−0.0449	0.0393	0.1574	−0.6934	1.3171	−1.3069	0.6799	−0.143	0
S5	0	−0.0513	−0.0193	−0.016	0.0043	0.0034	−0.0155	0.0319	−0.0128	0
S6	0	−0.1089	−0.0569	0.3576	−0.9255	1.1947	−0.8604	0.3322	−0.0547	0
S7	−7.5	−0.2139	−0.0107	0.1788	−0.1827	−0.1159	0.3046	−0.1897	0.0405	0
S8	−43.341	−0.1402	−0.061	0.2777	−0.4123	0.3523	−0.1857	0.0564	−0.0071	0
S9	−35.081	−0.0602	0.0736	−0.1046	0.1084	−0.0726	0.0255	−0.0041	0.0002	0
S10	−1.5734	0.1621	−0.2197	0.1896	−0.107	0.0396	−0.0091	0.0011	−6E−05	0
S11	0.5153	0.2137	−0.3167	0.2399	−0.1217	0.0384	−0.0069	0.0007	−3E−05	0
S12	−1.1466	0.1967	−0.2565	0.1542	−0.0532	0.0115	−0.0015	0.0001	−4E−06	0
S13	−0.9056	−0.0077	−0.2094	0.1883	−0.0749	0.0167	−0.0022	0.0002	−5E−06	0
S14	−1.2797	−0.2192	0.1006	−0.0338	0.0088	−0.0018	0.0003	−2E−05	1E−06	−3E−08

Twenty-Sixth Example

[0276] FIG. **51** is a view illustrating a twenty-sixth example of an optical imaging system, and FIG. **52** illustrates aberration curves of the optical imaging system of FIG. **51**.

[0277] An optical imaging system **26** includes a first lens **1026**, a second lens **2026**, a third lens **3026**, a fourth lens **4026**, a fifth lens **5026**, a sixth lens **6026**, and a seventh lens **7026**.

[0278] The first lens **1026** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2026** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3026** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens **4026** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5026** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6026** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6026**. The seventh lens **7026** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7026**, and one inflection point is formed on the image-side surface of the seventh lens **7026**.

[0279] The optical imaging system **26** further includes a stop, a filter **8026**, and an image sensor **9026**. The stop is disposed between the second lens **2026** and the third lens **3026** to adjust an amount of light incident on the image sensor **9026**. The filter **8026** is disposed between the seventh lens **7026** and the image sensor **9026** to block infrared rays. The image sensor **9026** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **51**, the stop is disposed at a distance of 0.8664 mm from the object-side surface of the first lens **1026** toward the imaging plane of the optical imaging system **26**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 26 listed in Table 57 that appears later in this application.

[0280] Table 51 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **51**, and Table 52 below shows aspherical surface coefficients of the lenses of FIG. **51**.

TABLE-US-00051 TABLE 51 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.8830 0.5872 1.546 56.114 1.050 S2 Lens 18.0733 0.0492 0.962 S3 Second 4.5995 0.2300 1.667 20.353 0.934 Lens S4 (Stop) 2.5464 0.3929 0.837 S5 Third −21.7546 0.2745 1.546 56.114 1.100 S6 Lens −13.5144 0.0611 1.106 S7 Fourth 25.3349 0.2655 1.546 56.114 1.200 S8 Lens 25.3360 0.3710 1.285 S9 Fifth 9.4682 0.3930 1.656 21.525 1.500 S10 Lens 5.1029 0.3790 1.754 S11 Sixth 6.4162 0.8885 1.546 56.114 2.041 S12 Lens 6.3521 0.0460 2.631 S13 Seventh 1.9665 0.8854 1.536 55.656 3.050 S14 Lens 1.7699 0.3098 3.456 S15 Filter Infinity 0.2100 1.518 64.197 3.768 S16 Infinity 0.6537 3.829 S17 Imaging Infinity −0.0037 4.129 Plane

TABLE-US-00052 TABLE 52 K A B C D E F G H J S1 −0.1525 0.0035 0.0054 −0.0238 0.0587 −0.0925 0.0808 −0.0376 0.0069 0 S2 −36.188 −0.0554 0.191 −0.4954 0.9092 −1.1194 0.849 −0.3546 0.0617 0 S3 −0.1164 −0.0883 0.2264 −0.5273 0.9947 −1.274 1.0104 −0.4343 0.076 0 S4 0.3326 −0.0462 0.097 −0.2316 0.5455 −0.848 0.7854 −0.3759 0.0708 0 S5 51.758 −0.0119 −0.0911 0.3617 −0.9067 1.3845 −1.3014 0.6835 −0.1493 0 S6 42.164 0.0924 −0.5269 1.3558 −2.2584 2.5093 −1.8107 0.7611 −0.139 0 S7 −4.7579 0.1336 −0.5938 1.261 −1.8115 1.7924 −1.1666 0.4427 −0.0728 0 S8 −3.4393 0.0471 −0.1842 0.2886 −0.3575 0.3273 −0.1971 0.067 −0.0093 0 S9 −8.5449 −0.0502 −0.0588 0.1599 −0.2027 0.1398 −0.0542 0.0105 −0.0007 0 S10 −18.064 −0.044 −0.0734 0.1425 −0.1303 0.0691 −0.0217 0.0038 −0.0003 0 S11 −4.6497 0.0633 −0.1193 0.0882 −0.0426 0.0135 −0.0028 0.0004 −2E−05 0 S12 −50 0.034 −0.0497 0.0246 −0.0072 0.0013 −0.0001 7E−06 −2E−07 0 S13 −2.4291 −0.1201 0.0167 0.0022 −0.0009 0.0001 −6E−06 1E−07 9E−10 0 S14 −1.0032 −0.1111 0.0248 −0.0032 −0.0001 0.0001 −2E−05 2E−06 −8E−08 1E−09

Twenty-Seventh Example

[0281] FIG. **53** is a view illustrating a twenty-seventh example of an optical imaging system, and FIG. **54** illustrates aberration curves of the optical imaging system of FIG. **53**.

[0282] An optical imaging system **27** includes a first lens **1027**, a second lens **2027**, a third lens **3027**, a fourth lens **4027**, a fifth lens **5027**, a sixth lens **6027**, and a seventh lens **7027**.

[0283] The first lens **1027** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2027** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3027** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens **4027** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5027** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6027** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6027**. The seventh lens **7027** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7027**, and one inflection point is formed on the image-side surface of the seventh lens **7027**.

[0284] The optical imaging system **27** further includes a stop, a filter **8027**, and an image sensor **9027**. The stop is disposed between the second lens **2027** and the third lens **3027** to adjust an amount of light incident on the image sensor **9027**. The filter **8027** is disposed between the seventh lens **7027** and the image sensor **9027** to block infrared rays. The image sensor **9027** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **53**, the stop is disposed at a distance of 0.9037 mm from the object-side surface of the first lens **1027** toward the imaging plane of the optical imaging system **27**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 27 listed in Table 57 that appears later in this application.

[0285] Table 53 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **53**, and Table 54 below shows aspherical surface coefficients of the lenses of FIG. **53**.

TABLE-US-00053 TABLE 53 Sur- Thick- Effective face Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.8987 0.6486 1.546 56.114 1.260 S2 Lens 7.3568 0.0250 1.216 S3 Second 3.8789 0.2300 1.667 20.353 1.161 S4 Lens 2.7620 0.3408 1.053 (Stop) S5 Third −50.1242 0.2819 1.546 56.114 1.120 S6 Lens −14.9889 0.0597 1.158 S7 Fourth 12.0498 0.2698 1.546 56.114 1.220 S8 Lens 12.5657 0.2919 1.320 S9 Fifth 9.5926 0.3500 1.667 20.353 1.520 S10 Lens 5.2748 0.3344 1.762 S11 Sixth 6.8735 0.8484 1.546 56.114 2.052 S12 Lens 7.4933 0.0591 2.641 S13 Seventh 2.0337 0.8836 1.536 55.656 3.070 S14 Lens 1.8436 0.3048 3.425 S15 Filter Infinity 0.2100 1.518 64.197 3.764 S16 Infinity 0.6441 3.825 S17 Imaging Infinity 0.0150 4.134 Plane

TABLE-US-00054 TABLE 54 K A B C D E F G H J S1 −0.1061 −0.0082 0.0469 −0.0925 0.0811 −0.0129 −0.032 0.0224 −0.0047 0 S2 −36.188 −0.0502 0.1624 −0.4029 0.6931 −0.7643 0.5021 −0.1789 0.0264 0 S3 0.0036 −0.0795 0.2057 −0.548 1.0742 −1.291 0.9097 −0.3412 0.052 0 S4 0.4038 −0.0325 0.0884 −0.3009 0.7004 −0.9194 0.6738 −0.2424 0.0308 0 S5 51.758 0.0055 −0.1746 0.5018 −0.9395 1.1442 −0.9144 0.4407 −0.0937 0 S6 42.164 0.0953 −0.4992 1.0397 −1.2284 0.8169 −0.2802 0.0384 4E−06 0 S7 −4.7579 0.1185 −0.4938 0.8554 −0.8643 0.5167 −0.185 0.0417 −0.0054 0 S8 −3.4393 0.0492 −0.194 0.3147 −0.3773 0.3249 −0.1878 0.063 −0.0088 0 S9 −8.5449 −0.0638 0.0289 −0.0884 0.1649 −0.171 0.0983 −0.0306 0.0041 0 S10 −18.064 −0.0543 −0.0172 0.0321 −0.0179 0.004 5E−06 −0.0001 8E−06 0 S11 −4.6497 0.0535 −0.0909 0.0613 −0.0311 0.011 −0.0026 0.0004 −2E−05 0 S12 −50 0.0103 −0.0176 0.0057 −0.0015 0.0003 −4E−05 2E−06 −6E−08 0 S13 −2.606 −0.1177 0.0192 −0.0004 −1E−04 −1E−05 4E−06 −4E−07 9E−09 0 S14 −1.0102 −0.0979 0.0187 −0.0024 0.0001 2E−05 −6E−06 6E−07 −3E−08 6E−10

Twenty-Eighth Example

[0286] FIG. **55** is a view illustrating a twenty-eighth example of an optical imaging system, and FIG. **56** illustrates aberration curves of the optical imaging system of FIG. **55**.

[0287] An optical imaging system **28** includes a first lens **1028**, a second lens **2028**, a third lens **3028**, a fourth lens **4028**, a fifth lens **5028**, a sixth lens **6028**, and a seventh lens **7028**.

[0288] The first lens **1028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3028** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5028** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6028** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6028**. The seventh lens **7028** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7028**.

[0289] The optical imaging system **28** further includes a stop, a filter **8028**, and an image sensor **9028**. The stop is disposed between the second lens **2028** and the third lens **3028** to adjust an amount of light incident on the image sensor **9028**. The filter **8028** is disposed between the seventh lens **7028** and the image sensor **9028** to block infrared rays. The image sensor **9028** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **55**, the stop is disposed at a distance of 1.2500 mm from the object-side surface of the first lens **1028** toward the imaging plane of the optical imaging system **28**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 28 listed in Table 57 that appears later in this application.

[0290] Table 55 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **55**, and Table 56 below shows aspherical surface coefficients of the lenses of FIG. **55**.

TABLE-US-00055 TABLE 55 Sur- Thick- Effective face Radius of ness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 2.2813 0.4941 1.546 56.114 1.640 S2 Lens 2.7190 0.1308 1.629 S3 Second 2.7090 0.6000 1.546 56.114 1.595 S4 Lens 37.2451 0.0250 1.593 S5 (Stop) 11.0960 0.2300 1.679 19.236 1.522 S6 Third 4.3260 0.4820 1.450 Lens S7 Fourth 8.7091 0.3517 1.656 21.525 1.514 S8 Lens 14.9889 0.4926 1.555 S9 Fifth 5.3406 0.4257 1.679 19.236 1.840 S10 Lens 2.9721 0.1952 2.415 S11 Sixth 2.1211 0.5425 1.546 56.114 2.362 S12 Lens 4.9818 0.4212 2.717 S13 Seventh 2.4999 0.5046 1.546 56.114 3.169 S14 Lens 1.4794 0.2230 3.372 S15 Filter Infinity 0.2100 1.518 64.197 3.779 S16 Infinity 0.6566 3.848 S17 Imaging Infinity 0.0150 4.210 Plane

TABLE-US-00056 TABLE 56 K A B C D E F G H J S1 −0.9867 −0.0114 0.0111 −0.0538 0.0917 −0.0925 0.0542 −0.0183 0.0033 −0.0003 S2 −12.035 0.0479 −0.173 0.2637 −0.3194 0.2597 −0.1301 0.0386 −0.0062 0.0004 S3 −0.9455 −0.0107 −0.0528 0.1054 −0.1922 0.2187 −0.1388 0.0499 −0.0096 0.0008 S4 3.0384 −0.0418 0.2162 −0.5714 0.8143 −0.6941 0.3642 −0.1154 0.0202 −0.0015 S5 10.164 −0.0842 0.2583 −0.5961 0.8439 −0.7413 0.4043 −0.1328 0.0241 −0.0018 S6 2.0809 −0.0648 0.1424 −0.2905 0.4095 −0.3776 0.2224 −0.0802 0.0163 −0.0014 S7 −13.097 −0.0215 −0.0518 0.145 −0.2583 0.2777 −0.1843 0.0733 −0.0157 0.0014 S8 5.8592 −0.0435 0.0377 −0.0856 0.1074 −0.0887 0.0479 −0.0161 0.0031 −0.0003 S9 −43.521 −0.0279 0.0228 0.0104 −0.0468 0.0468 −0.0253 0.0078 −0.0013 9E−05 S10 −17.628 −0.0671 0.0426 −0.0048 −0.0103 0.0068 −0.0022 0.0004 −4E−05 1E−06 S11 −9.8081 0.0426 −0.1025 0.0919 −0.0514 0.0177 −0.0039 0.0006 −5E−05 2E−06 S12 −0.0695 0.0048 −0.0496 0.0379 −0.0164 0.004 −0.0005 4E−05 −8E−07 −1E−08 S13 −0.6908 −0.2261 0.0409 0.0129 −0.0077 0.0017 −0.0002 1E−05 −6E−07 1E−08 S14 −1.419 −0.2123 0.0904 −0.0281 0.0063 −0.001 9E−05 −6E−06 2E−07 −2E−09

[0291] Table 57 below shows an overall focal length f of the optical imaging system, an overall length TTL of the optical imaging system (a distance from the object-side surface of the first lens to the imaging plane), a distance SL from the stop to the imaging plane, an f-number (F No.) of the optical imaging system (the overall focal length f of the optical imaging system divided by the diameter of an entrance pupil of the optical imaging system, where both f and the diameter of the entrance pupil are expressed in mm), an image height (IMG HT) on the imaging plane (one-half of a diagonal length of the imaging plane), and a field of view (FOV) of the optical imaging system for each of Examples 1-28 described herein. The values of f, TTL, SL, and IMG HT are expressed

in mm. The values of F No. are dimensionless values. The values of FOV are expressed in degrees.
TABLE-US-00057 TABLE 57 IMG Example f TTL SL F No. HT FOV f134567 1 4.8480 6.000 5.0965 1.540 4.000 77.800 3.4441 2 4.2829 5.1899 3.9385 1.5853 3.6900 79.815 −23.2030 3 4.7800 5.8272 4.4022 1.5729 4.0500 79.019 −42.8744 4 3.9500 4.8189 3.6503 1.5808 3.2500 77.470 −30.5480 5 4.3500 5.3000 4.9170 1.5800 3.3844 79.580 −5.2273 6 4.2800 5.1000 4.3688 1.7100 3.5352 77.840 2.7518 7 4.4011 5.3000 4.1423 1.6897 3.7280 79.310 −26.4128 8 4.5110 5.4999 4.4199 1.6585 3.7280 77.849 392.6438 9 4.5442 5.5000 4.4233 1.6724 3.7280 77.539 −58.4566 10 4.8227 5.8648 4.6866 1.6639 3.7280 73.384 89.9938 11 4.5369 5.5000 4.2704 1.5688 3.7280 77.565 −21.5648 12 4.8309 5.9552 4.7759 1.6624 4.1280 79.742 −11.9581 13 5.0858 5.9999 4.7947 1.5909 4.1280 76.863 −8.3987 14 4.4468 5.1439 4.8939 2.0717 3.5280 75.627 2.5748 15 4.4000 5.2000 1.8078 3.2610 72.552 3.0701 16 3.9935 5.1246 4.4837 1.5723 3.2610 77.383 3.8824 17 3.9200 4.7000 4.0189 1.8300 3.2610 78.330 2.7493 18 4.0052 4.9400 3.8888 1.5800 3.2260 76.500 6.5989 19 4.3329 5.3200 4.9460 1.4980 3.7520 80.300 −5.0314 20 4.7371 5.8000 4.7399 1.5686 4.2500 82.209 3.7697 21 4.5877 5.6173 4.4893 1.6867 3.7280 76.901 −157.3713 22 4.4980 5.5000 4.2749 1.6297 3.7280 78.012 −9.9003 23 4.5861 5.4613 4.5098 1.7914 3.7280 76.963 3.1230 24 4.3158 5.2500 4.3926 1.6913 3.7280 80.429 3.0710 25 4.3017 5.2400 4.3678 1.9545 3.7280 80.465 3.0059 26 4.9659 5.9933 5.1269 2.3647 4.1280 78.448 3.1668 27 4.6674 5.7971 4.8934 1.8451 4.1280 81.802 3.6413 28 4.8700 6.0000 4.7500 1.4848 4.2000 80.288 −23.7892

[0292] Table 58 below shows in mm a focal length f1 of the first lens, a focal length f2 of the second lens, a focal length f3 of the third lens, a focal length f4 of the fourth lens, a focal length f5 of the fifth lens, a focal length f6 of the sixth lens, and a focal length f7 of the seventh lens for each of Examples 1-28 described herein.

TABLE-US-00058 TABLE 58 Example f1 f2 f3 f4 f5 f6 f7 1 4.699 −11.772 21.070 −38.885 −12.922 2.771 −2.453 2 9.5890 4.5406 −6.9872 17555492 558.7000 −30.322 1998.195 3 10.0345 5.2922 −7.6128 22346862 86.5836 −27.728 146.074 4 8.4094 4.3550 −6.5204 −4512 74.3686 −22.452 1842.731 5 −64.2326 3.2480 −7.4275 −43.7223 52.4247 3.010 −2.424 6 3.5960 −7.3490 −1245.24 15.6567 −19.7232 2.662 −2.171 7 9.9516 4.9854 −9.0419 −60.9593 28.4615 −19.130 −36.205 8 12.2172 6.0167 −9.9247 25.3582 28.7232 −32.884 −16.737 9 13.4191 5.6271 −8.9208 16.1417 −36.7578 29.873 −12.281 10 13.6636 6.5561 −11.4351 32.2946 28.3279 −36.885 −22.967 11 17.0122 4.9962 −9.0344 25.7533 24.3015 −79.899 −10.773 12 32.0696 4.9582 −9.5104 22.5720 16.1804 −12.867 −20.135 13 43.5223 4.4621 −8.8732 52.4923 16.2883 −17.315 −19.385 14 3.6264 −6.9779 10.5508 125.3810 −28.1554 −367.720 −9.031 15 4.2900 −10.6063 30.9779 14.8711 −21.1331 3.784 −2.465 16 5.6767 −73.5511 −122.7160 15.5097 207.3750 3.799 −2.466 17 3.5400 −8.7600 −87.3600 14.1800 −64.1800 −799.990 −18.040 18 5.0178 11.6357 −8.1681 −18.7682 18.1322 2.601 −2.226 19 −31.5304 3.1365 −7.5452 −130.0329 80.8864 2.966 −2.423 20 5.1067 −17.5830 27.4583 −30.2565 −75.8106 2.526 −2.062 21 14.2697 5.4868 −9.0060 21.0725 −18.2040 43.002 92.362 22 20.3702 4.0105 −7.6691 107.4534 −23.0047 189.703 54.850 23 3.8083 −9.4079 −530.7502 22.8366 −27.1051 −22.324 66.015 24 4.2066 −10.3310 27.6314 107.6478 4.1658 7.509 −2.062 25 3.6197 −10.4284 39.8209 −38.7622 4.3417 10.303 −2.323 26 3.8015 −8.9549 64.5946 12384.8 −17.5030 299.093 57.797 27 4.4990 −15.6740 39.0579 453.7793 −18.1602 102.612 59.134 28 18.5366 5.3131 −10.5800 30.6731 −10.6364 6.334 −8.036

[0293] Table 59 below shows in mm a thickness (L1edgeT) of an edge of the first lens, a thickness (L2edgeT) of an edge of the second lens, a thickness (L3edgeT) of an edge of the third lens, a thickness (L4edgeT) of an edge of the fourth lens, a thickness (L5edgeT) of an edge of the fifth lens, a thickness (L6edgeT) of an edge of the sixth lens, and a thickness (L7edgeT) of an edge of the seventh lens for each of Examples 1-28 described herein.

TABLE-US-00059 TABLE 59 Example L1edgeT L2edgeT L3edgeT L4edgeT L5edgeT L6edgeT L7edgeT 1 0.2527 0.3544 0.2546 0.3479 0.2843 0.2558 0.8308 2 0.2513 0.2829 0.3599 0.2195 0.2899 0.3277 0.3914 3 0.2822 0.3127 0.4160 0.2370 0.3248 0.3709 0.3823 4 0.2327 0.2588

0.3335 0.2027 0.2718 0.3298 0.3765 5 0.2200 0.2700 0.3480 0.2240 0.2590 0.2690 0.4370 6 0.2216 0.3773 0.2347 0.2401 0.1894 0.2600 0.3234 7 0.2568 0.2552 0.3401 0.2756 0.3650 0.3065 0.2776 8 0.2535 0.2550 0.4864 0.2519 0.3609 0.4901 0.4086 9 0.2497 0.2500 0.4399 0.2704 0.3183 0.6519 0.2941 10 0.2855 0.3362 0.4980 0.2610 0.4273 0.4966 0.4889 11 0.2582 0.2578 0.4040 0.2530 0.4107 0.2818 0.4636 12 0.2479 0.3333 0.3826 0.3612 0.5824 0.3553 0.6406 13 0.2542 0.1246 0.4252 0.2768 0.3797 0.4611 0.3277 14 0.2688 0.3078 0.1901 0.2300 0.4099 0.7139 0.3000 15 0.2048 0.4069 0.2010 0.3332 0.2778 0.3483 0.8151 16 0.2180 0.3468 0.2110 0.2593 0.2768 0.2512 0.9497 17 0.1000 0.2800 0.1200 0.4200 0.1600 0.4100 0.5800 18 0.2320 0.2180 0.3500 0.2220 0.2410 0.3730 0.3970 19 0.2203 0.2484 0.3500 0.2373 0.2517 0.2418 0.5401 20 0.4351 0.3299 0.2145 0.3306 0.1429 0.2681 0.5558 21 0.2502 0.3421 0.3843 0.4086 0.2945 0.7273 0.2827 22 0.2675 0.3957 0.3836 0.4004 0.2278 0.5887 0.3421 23 0.2307 0.2892 0.2545 0.2544 0.3605 0.4762 0.6576 24 0.2501 0.3506 0.2323 0.3110 0.3641 0.3267 0.3719 25 0.2520 0.2935 0.2377 0.3745 0.2580 0.4152 0.6857 26 0.2927 0.2979 0.2516 0.2513 0.4092 0.7155 0.6778 27 0.2463 0.2800 0.2542 0.2728 0.3562 0.6300 0.6917 28 0.2515 0.1596 0.4288 0.3024 0.4301 0.3912 0.4326

[0294] Table 60 below shows in mm a sag (L5S1 sag) of an object-side surface of the fifth lens, a sag (L5S2 sag) of an image-side surface of the fifth lens, a thickness (Yc71P1) of the seventh lens at a first inflection point on the object-side surface of the seventh lens, a thickness (Yc71P2) of the seventh lens at a second inflection point on the object-side surface of the seventh lens, a thickness (Yc72P1) of the seventh lens at a first inflection point on the image-side surface of the seventh lens, and a thickness (Yc72P2) of the seventh lens at a second inflection point on the image-side surface of the seventh lens for each of Examples 1-28 described herein.

TABLE-US-00060 TABLE 60 Exam- L5S2 ple L5S1 sag sag Yc71P1 Yc71P2 Yc72P1 Yc72P2 1 −0.4639 −0.4961 1.3100 — 0.9900 — 2 0.1069 0.1580 0.5970 0.6980 0.6920 — 3 0.1973 0.1988 0.6780 0.8030 0.7950 — 4 0.2004 0.2021 0.5680 0.6700 0.6670 — 5 0.1154 0.1393 0.9300 — 0.8110 — 6 −0.4658 −0.5261 2.9330 — 4.1420 — 7 0.2103 0.2454 0.5690 0.6410 0.6700 — 8 0.1024 0.1416 0.5620 — 0.6860 — 9 0.1850 0.2667 0.5270 0.4850 0.6470 — 10 0.0544 0.0531 0.6070 — 0.7170 — 11 0.1496 0.0888 0.5070 0.7380 0.6370 — 12 0.1290 0.0520 0.6280 0.9230 0.7900 — 13 0.0698 0.0604 0.6330 0.5190 0.7380 — 14 −0.2605 −0.2625 0.4730 — 0.6310 — 15 −0.4848 −0.4070 0.8900 — 0.9200 — 16 −0.4791 −0.4221 — — 0.7810 — 17 −0.4400 −0.4500 0.7300 — 1.1200 — 18 −0.3405 −0.5395 0.8247 — 0.7357 — 19 0.2023 0.2006 0.9670 — 0.5350 0.9040 20 0.3540 0.4306 1.0040 — 0.5110 0.9130 21 0.2211 0.3179 0.5700 0.4520 0.6330 — 22 0.1756 0.3016 0.5950 — 0.6720 — 23 0.2805 0.2808 0.8830 0.9150 0.9880 — 24 0.4949 0.7331 — — 0.7340 — 25 0.2760 0.5093 — — 0.9680 — 26 0.0918 0.1026 0.9550 1.1030 1.1280 — 27 0.1793 0.1731 0.9640 1.1140 1.1300 — 28 0.2606 0.2562 0.5870 — 0.7630 —

[0295] Table 61 below shows in mm an inner diameter of each of the first to seventh spacers for each of Examples 1-28 described herein. S1d is an inner diameter of the first spacer SP1, S2d is an inner diameter of the second spacer SP2, S3d is an inner diameter of the third spacer SP3, S4d is an inner diameter of the fourth spacer SP4, S5d is an inner diameter of the fifth spacer SP5, S6d is an inner diameter of the sixth spacer SP6, and S7d is an inner diameter of the seventh spacer SP7.

TABLE-US-00061 TABLE 61 Ex- am- ple S1d S2d S3d S4d S5d S6d S7d 1 2.8400 2.5300 2.8300 3.2900 4.3400 6.3100 — 2 1.3500 1.2300 1.1400 1.5300 2.0700 2.7800 — 3 1.5000 1.3400 1.3200 1.7200 2.3100 3.0300 — 4 1.2400 1.1500 1.0300 1.4800 1.9000 2.4600 — 5 1.3400 1.2300 1.0300 1.5000 1.9800 2.6600 — 6 2.3100 2.1600 2.5400 2.9400 4.0600 4.8400 5.12 7 2.5800 2.4000 2.4900 2.9700 4.1600 4.8900 5.51 8 2.5900 2.5000 2.5300 2.9000 3.8000 4.9000 — 9 2.6500 2.4600 2.3900 2.9000 3.8000 5.1500 — 10 2.7700 2.6100 2.7900 3.1200 4.0300 4.8900 — 11 2.8100 2.6300 2.6500 3.1200 4.0300 4.9100 — 12 2.8000 2.6500 2.7300 3.5400 3.4200 4.4400 5.74 13 3.1700 3.0000 2.7900 3.0800 4.1800 5.4900 — 14 2.1200 2.1000 2.0400 2.1200 2.8100 4.6400 — 15 2.3200 2.3600 2.5600 2.9300 3.7000 4.3500 — 16 2.4100 2.3000 2.6600 3.0300 3.7600 — — 17 2.1060 1.8860 2.0080 2.7000 3.0740 4.4840 — 18 2.4200 2.2300 2.0900 2.4700

3.2000 4.3300 — 19 2.8800 2.6300 2.2900 2.9300 4.3800 5.5100 — 20 2.6600 2.4700 X 3.1300 3.7800 5.1300 — 21 2.6700 2.5000 2.4400 2.9900 3.8000 5.2700 — 22 2.7100 2.5300 2.5200 3.0300 3.7800 4.8300 — 23 2.3600 2.0300 2.2500 2.6500 3.6400 5.1400 5.3 24 2.3300 2.2700 2.5300 3.1700 4.5200 5.3100 5.64 25 2.0600 1.8900 2.1500 2.7000 3.6100 4.5600 4.84 26 1.8900 1.8400 2.3300 2.7300 3.7300 5.4300 6.03 27 2.3900 2.1500 2.4000 2.8200 3.9400 5.6800 6.02 28 3.2200 3.1100 2.9200 3.2500 4.6000 5.6000 6.15

[0296] Table 62 below shows in mm.sup.3 a volume of each of the first to seventh lenses for each of Examples 1-28 described herein. L1v is a volume of the first lens, L2v is a volume of the second lens, L3v is a volume of the third lens, L4v is a volume of the fourth lens, L5v is a volume of the fifth lens, L6v is a volume of the sixth lens, and L7v is a volume of the seventh lens.

TABLE-US-00062 TABLE 62 Ex- am- ple L1v L2v L3v L4v L5v L6v L7v 1 9.4074 6.0389 7.8806 9.9733 14.2491 18.7217 48.3733 2 6.5805 7.1213 7.7660 6.6371 11.7744 12.5638 20.4308 3 8.0184 9.5628 9.6052 8.4128 12.0326 16.7196 28.0267 4 6.3442 6.9494 7.7597 6.2076 6.8959 10.3364 16.5597 5 5.7249 8.0179 8.3774 7.9589 10.3434 11.1031 27.1511 6 5.2342 5.0595 5.1455 4.1402 5.9856 8.1378 19.6812 7 5.6390 4.8580 6.6748 7.1627 11.0369 11.9357 27.1217 8 5.1778 5.1427 8.2986 6.3777 12.6369 14.9811 20.7310 9 5.1650 5.3015 6.2461 7.0472 12.2503 19.1335 17.9152 10 5.9227 6.9971 9.1275 7.0250 12.2307 15.4792 23.2093 11 6.0930 6.3798 6.8569 7.4035 9.7509 11.7344 23.4758 12 5.1989 7.9980 6.6804 7.9875 20.4160 20.0925 36.0790 13 6.7496 7.2862 8.8499 8.0745 13.4156 20.3714 25.3575 14 3.8115 4.6714 4.0552 5.0631 11.2844 25.7618 16.5646 15 4.2347 5.5368 5.5931 7.5471 9.4202 8.9992 27.3258 16 4.6529 4.6572 6.2312 6.7131 10.2673 11.7401 33.5372 17 2.5134 3.7749 2.3033 9.4226 4.0073 16.0487 22.2874 18 4.3198 3.6956 4.1821 5.1874 8.1714 10.5471 19.1646 19 5.6174 7.9604 6.8464 7.2237 12.5253 12.8147 28.5967 20 11.1287 6.0119 5.8018 9.6434 7.9251 20.0604 37.3066 21 5.0360 6.7314 5.9764 9.3728 10.4859 21.6926 17.1978 22 4.9598 8.1220 7.2222 8.2929 8.6024 18.8370 19.7464 23 5.4854 3.9796 4.1274 4.6927 9.8848 20.3357 35.3318 24 5.5446 5.0525 4.5199 5.6552 9.8279 14.9067 22.4415 25 3.8100 3.9751 3.9272 6.1885 7.5160 13.0347 31.8586 26 4.7517 4.3655 6.4562 5.0723 9.8674 36.8705 47.4701 27 5.6273 4.9490 5.1423 5.0791 9.3624 31.5832 47.9081 28 7.5192 7.1322 12.3605 9.0385 17.2925 20.5539 33.5701

[0297] Table 63 below shows in mg a weight of each of the first to seventh lenses for each of Examples 1-28 described herein. L1w is a weight of the first lens, L2w is a weight of the second lens, L3w is a weight of the third lens, L4w is a weight of the fourth lens, L5w is a weight of the fifth lens, Low is a weight of the sixth lens, and L7w is a weight of the seventh lens.

TABLE-US-00063 TABLE 63 Ex- am- ple L1w L2w L3w L4w L5w L6w L7w 1 9.7837 7.4278 8.1958 12.2672 17.8114 19.4706 50.3082 2 6.8437 7.4062 9.7075 8.2964 12.2454 15.7048 20.6351 3 8.3391 9.9453 12.0065 10.5160 12.5139 20.8995 28.3070 4 6.5980 7.2274 9.6996 7.7595 7.1717 12.9205 16.7253 5 5.9539 8.3386 10.4718 9.7099 12.6189 11.5472 28.2371 6 5.4436 6.2232 5.3513 4.3058 7.3623 8.4633 20.4684 7 5.8646 5.0523 8.3435 8.9534 11.4784 14.9196 27.3929 8 5.3849 5.3484 10.3733 6.6328 13.1424 15.5803 21.5602 9 5.3716 5.5136 7.8076 7.3291 15.3129 19.8988 18.6318 10 6.1596 7.2770 11.4094 7.3060 12.7199 19.3490 24.1377 11 6.3367 6.6350 8.5711 7.6996 10.1409 14.6680 24.4148 12 5.4069 8.3179 8.3505 8.3070 21.2326 25.1156 37.5222 13 7.0196 7.5776 11.0624 10.0931 13.9522 25.4643 26.3718 14 3.9640 5.7458 4.2174 5.2656 14.1055 26.7923 17.2272 15 4.4041 6.8103 5.8168 7.8490 11.5868 9.3592 28.4188 16 4.8390 5.7284 6.4804 6.9816 12.6288 12.2097 34.8787 17 2.6139 4.6431 2.8331 9.7995 5.0091 20.0609 22.5103 18 4.4926 3.8434 5.1440 5.3949 10.2143 10.9690 19.9312 19 5.8421 8.2788 8.5580 9.0296 15.6566 13.3273 29.7406 20 11.5738 7.5149 6.0339 11.8614 9.6686 20.8628 38.7989 21 5.2374 7.0007 7.4705 9.7477 13.1074 22.5603 17.8857 22 5.1582 8.4469 9.0278 10.3661 8.9465 19.5905 20.5363 23 5.7048 4.8949 5.0767 4.8804 12.3560 25.4196 35.6851 24 5.7664 6.3156 4.7007 7.0690 10.2210 15.5030 23.3392 25 3.9624 4.8894 4.0843 7.6119 7.8166 13.5561 33.1329 26

4.9418 5.3696 6.7144 5.2752 12.3343 38.3453 47.9448 27 5.8524 6.0873 5.3480 5.2823 11.5158 32.8465 48.3872 28 7.8200 7.4175 15.4506 11.2981 21.6156 21.3761 34.9129

[0298] Table 64 below shows in mm an overall outer diameter (including a rib) of each of the first to seventh lenses for each of Examples 1-28 described herein. L1TR is an overall outer diameter of the first lens, L2TR is an overall outer diameter of the second lens, L3TR is an overall outer diameter of the third lens, L4TR is an overall outer diameter of the fourth lens, L5TR is an overall outer diameter of the fifth lens, L6TR is an overall outer diameter of the sixth lens, and L7TR is an overall outer diameter of the seventh lens.

TABLE-US-00064 TABLE 64

Example	L1TR	L2TR	L3TR	L4TR	L5TR	L6TR	L7TR
1	4.930	5.130	5.630	6.230	7.200	7.600	7.800
2	2.270	2.390	2.520	2.750	3.020	3.210	3.320
3	2.440	2.540	2.690	2.900	3.190	3.440	3.620
4	2.290	2.400	2.540	2.630	2.780	2.910	3.040
5	2.460	2.580	2.690	2.800	3.170	3.310	3.470
6	4.220	4.420	4.540	4.720	5.400	5.740	6.300
7	4.210	4.300	4.440	4.840	5.470	6.120	6.900
8	4.250	4.340	4.480	4.880	5.510	6.160	6.480
9	4.190	4.280	4.410	4.810	5.510	6.160	6.520
10	4.430	4.520	4.660	5.060	5.500	6.260	6.580
11	4.430	4.520	4.660	5.060	5.500	6.260	6.570
12	4.470	4.560	4.700	5.030	6.660	7.180	7.430
13	4.730	4.820	4.960	5.290	5.880	6.810	7.060
14	3.510	3.810	4.390	4.980	5.850	6.150	6.250
15	3.930	4.130	4.710	6.170	5.300	6.570	6.670
16	4.030	4.230	4.810	5.400	6.270	6.670	6.770
17	3.830	4.078	4.220	4.980	5.740	6.174	6.510
18	3.930	4.130	4.330	4.930	5.420	5.820	6.020
19	4.630	4.830	5.030	5.830	6.320	6.720	6.920
20	4.830	5.030	5.230	6.030	6.520	6.920	7.120
21	4.250	4.340	4.480	4.880	5.510	6.330	6.700
22	4.290	4.380	4.520	4.920	5.650	6.260	6.770
23	4.090	4.180	4.300	4.530	5.220	6.620	7.320
24	4.110	4.200	4.340	4.612	5.550	6.350	7.210
25	3.730	3.820	3.960	4.390	4.960	6.000	6.860
26	3.970	4.060	4.190	4.630	5.200	7.150	8.020
27	4.390	4.480	4.610	5.040	5.610	7.090	7.950
28	4.870	4.960	5.090	5.520	6.370	7.410	7.840

[0299] Table 65 below shows in mm a thickness of a flat portion of the rib of each of the first to seventh lenses for each of Examples 1-28 described herein. L1rt is a thickness of a flat portion of the rib of the first lens, L2rt is a thickness of a flat portion of the rib of the second lens, L3rt is a thickness of a flat portion of the rib of the third lens, L4rt is a thickness of a flat portion of the rib of the fourth lens, L5rt is a thickness of a flat portion of the rib of the fifth lens, L6rt is a thickness of a flat portion of the rib of the sixth lens, and L7rt is a thickness of a flat portion of the rib of the seventh lens.

TABLE-US-00065 TABLE 65

Example	L1rt	L2rt	L3rt	L4rt	L5rt	L6rt	L7rt
1	0.580	0.380	0.330	0.300	0.390	0.475	0.920
2	0.590	0.480	0.510	0.270	0.480	0.310	0.410
3	0.600	0.580	0.560	0.470	0.340	0.400	0.470
4	0.540	0.500	0.520	0.420	0.210	0.390	0.400
5	0.390	0.440	0.470	0.360	0.420	0.380	0.470
6	0.435	0.430	0.360	0.215	0.320	0.330	0.405
7	0.550	0.380	0.580	0.410	0.500	0.320	0.530
8	0.490	0.390	0.680	0.360	0.630	0.490	0.480
9	0.520	0.420	0.520	0.410	0.610	0.700	0.370
10	0.490	0.470	0.710	0.470	0.560	0.500	0.550
11	0.560	0.430	0.560	0.510	0.400	0.350	0.550
12	0.490	0.530	0.520	0.370	0.620	0.520	0.690
13	0.600	0.440	0.600	0.460	0.580	0.600	0.440
14	0.482	0.395	0.316	0.328	0.422	0.885	0.409
15	0.431	0.556	0.361	0.429	0.380	0.380	0.667
16	0.431	0.457	0.361	0.364	0.380	0.334	0.729
17	0.326	0.433	0.265	0.472	0.156	0.520	0.641
18	0.440	0.330	0.300	0.260	0.425	0.500	0.518
19	0.540	0.480	0.460	0.250	0.555	0.395	0.688
20	0.620	0.400	0.270	0.430	0.300	0.520	0.760
21	0.480	0.490	0.480	0.500	0.470	0.830	0.320
22	0.480	0.560	0.580	0.470	0.340	0.620	0.440
23	0.510	0.250	0.320	0.320	0.510	0.530	0.720
24	0.510	0.450	0.340	0.430	0.410	0.410	0.420
25	0.400	0.420	0.370	0.500	0.320	0.460	0.720
26	0.470	0.410	0.450	0.410	0.470	0.930	0.700
27	0.440	0.390	0.400	0.400	0.380	0.740	0.720
28	0.560	0.410	0.560	0.540	0.520	0.440	0.540

[0300] Table 66 below shows, for each of Examples 1-28 described herein, dimensionless values of each of the ratio L1w/L7w in Conditional Expressions 1 and 7, the ratio S6d/f in Conditional Expressions 2 and 8, the ratio L1TR/L7TR in Conditional Expressions 3 and 9, the ratio L1234TRavg/L7TR in Conditional Expressions 4 and 10, the ratio L12345TRavg/L7TR in Conditional Expressions 5 and 11, and the ratio |f134567−f|/f in Conditional Expressions 6 and 12. The dimensionless value of each of these ratios is obtained by dividing two values expressed in a same unit of measurement.

TABLE-US-00066 TABLE 66

Example	L1w/L7w	S6d/f	L1TR/L7TR	L1234TRavg/L7TR	L12345TRavg/L7TR	|f134567-f|/f
1	0.1945	1.3016	0.6321	0.703	0.747	0.290
2	0.3317	0.6491	0.6837	0.748	0.780	6.418
3	0.2946	0.6339	0.6740	0.730	0.760	9.970
4	0.3945	0.6228	0.7533	0.811	0.832	8.734
5	0.2109	0.6115	0.7089	0.759	0.790	2.202
6	0.2660	1.1308	0.6698	0.710	0.740	0.357
7	0.2141	1.1111	0.6101	0.645	0.674	7.001
8	0.2498	1.0862	0.6559	0.693	0.724	86.041
9	0.2883	1.1333	0.6426	0.678	0.712	13.864
10	0.2552	1.0140	0.6733	0.709	0.735	17.660
11	0.2595	1.0822	0.6743	0.710	0.736	5.753
12	0.1441	0.9191	0.6016	0.631	0.684	3.475
13	0.2662	1.0795	0.6700	0.701	0.727	2.651
14	0.2301	1.0434	0.5616	0.668	0.721	0.421
15	0.1550	0.9886	0.5892	0.710	0.727	0.302
16	0.1387	0.5953	0.682	0.731	0.028	
17	0.1161	1.1439	0.5883	0.657	0.702	0.299
18	0.2254	1.0811	0.6528	0.719	0.755	0.648
19	0.1964	1.2717	0.6691	0.734	0.770	2.161
20	0.2983	1.0829	0.6784	0.742	0.776	0.204
21	0.2928	1.1487	0.6343	0.670	0.700	35.303
22	0.2512	1.0738	0.6337	0.669	0.702	3.201
23	0.1599	1.1208	0.5587	0.584	0.610	0.319
24	0.2471	1.2304	0.5700	0.599	0.633	0.288
25	0.1196	1.0600	0.5437	0.579	0.608	0.301
26	0.1031	1.0935	0.4950	0.525	0.550	0.362
27	0.1209	1.2170	0.5522	0.582	0.607	0.220
28	0.2240	1.1499	0.6212	0.652	0.684	5.885

[0301] FIGS. **57** and **58** are cross-sectional views illustrating examples of an optical imaging system and a lens barrel coupled to each other.

[0302] The examples of the optical imaging system **100** described in this application may include a self-alignment structure as illustrated in FIGS. **57** and **58**.

[0303] In one example illustrated in FIG. **57**, the optical imaging system **100** includes a self-alignment structure in which optical axes of four consecutive lenses **1000**, **2000**, **3000**, and **4000** are aligned with an optical axis of the optical imaging system **100** by coupling the four lenses **1000**, **2000**, **3000**, and **4000** to one another.

[0304] The first lens **1000** disposed closest to the object side of the optical imaging system **100** is disposed in contact with an inner surface of a lens barrel **200** to align the optical axis of the first lens **1000** with the optical axis of the optical imaging system **100**, the second lens **2000** is coupled to the first lens **1000** to align the optical axis of the second lens **2000** with the optical axis of the optical imaging system **100**, the third lens **3000** is coupled to the second lens **2000** to align the optical axis of the third lens **3000** with the optical axis of the optical imaging system **100**, and the fourth lens **4000** is coupled to the third lens **3000** to align the optical axis of the fourth lens **4000** with the optical axis of the optical imaging system **100**. The second lens **2000** to the fourth lens **4000** may not be disposed in contact with the inner surface of the lens barrel **200**.

[0305] Although FIG. **57** illustrates that the first lens **1000** to the fourth lens **4000** are coupled to one another, the four consecutive lenses that are coupled to one another may be changed to the second lens **2000** to a fifth lens **5000**, or the third lens **3000** to a sixth lens **6000**, or the fourth lens **4000** to a seventh lens **7000**.

[0306] In another example illustrated in FIG. **58**, the optical imaging system **100** includes a self-alignment structure in which optical axes of five consecutive lenses **1000**, **2000**, **3000**, **4000**, and **5000** are aligned with an optical axis of the optical imaging system **100** by coupling the five lenses **1000**, **2000**, **3000**, **4000**, and **5000** to one another.

[0307] The first lens **1000** disposed closest to the object side of the optical imaging system **100** is disposed in contact with an inner surface of the lens barrel **200** to align the optical axis of the first lens **1000** with the optical axis of the optical imaging system **100**, the second lens **2000** is coupled to the first lens **1000** to align the optical axis of the second lens **2000** with the optical axis of the optical imaging system **100**, the third lens **3000** is coupled to the second lens **2000** to align the optical axis of the third lens **3000** with the optical axis of the optical imaging system **100**, the fourth lens **4000** is coupled to the third lens **3000** to align the optical axis of the fourth lens **4000** with the optical axis of the optical imaging system **100**, and the fifth lens **5000** is coupled to the fourth lens **4000** to align the optical axis of the fifth lens **5000** with the optical axis of the optical imaging system **100**. The second lens **2000** to the fifth lens **5000** may not be disposed in contact with the inner surface of the lens barrel **200**.

[0308] Although FIG. **58** illustrates that the first lens **1000** to the fifth lens **5000** are coupled to one another, the five consecutive lenses that are coupled to one another may be changed to the second lens **2000** to a sixth lens **6000**, or the third lens **3000** to a seventh lens **7000**.

[0309] FIG. **59** is a cross-sectional view illustrating an example of a seventh lens.

[0310] FIG. **59** illustrates the overall outer diameter (L7TR) of the seventh lens, the thickness (L7rt) of the flat portion of the rib of the seventh lens, the thickness (L7edgeT) of the edge of the seventh lens, the thickness (Yc71P1) of the seventh lens at the first inflection point on the object-side surface of the seventh lens, the thickness (Yc71P2) of the seventh lens at the second inflection point on the object-side surface of the seventh lens, and the thickness (Yc72P1) of the seventh lens at the first inflection point on the image-side surface of the seventh lens. Although not illustrated in FIG. **59**, the seventh lens may also have a second inflection point on the image-side surface of the seventh lens, and a thickness of the seventh lens at this inflection point is Yc72P2 as listed in Table 60.

[0311] FIG. **60** is a cross-sectional view illustrating an example of a shape of a rib of a lens.

[0312] The examples of the optical imaging system **100** described in this application may include a structure for preventing a flare phenomenon and reflection.

[0313] For example, the ribs of the first to seventh lenses **1000**, **2000**, **3000**, **4000**, **5000**, **6000**, and **7000** of the optical imaging system may be partially surface-treated to make the surface of the rib rough as illustrated in FIG. **60**. Methods of surface treatment may include chemical etching, physical grinding, or any other surface treatment method capable of increasing a roughness of a surface.

[0314] A surface-treated area EA may be formed in an entire area from an edge of the optical portion of the lens through which light actually passes to an outer end of the rib. However, as illustrated in FIG. **60**, non-treated areas NEA including step portions E11, E21, and E22 may not be surface-treated, or may be surface-treated to have a roughness less than a roughness of the surface-treated area EA. The step portions E11, E21, and E22 are portions where the thickness of the rib abruptly changes. A width G1 of a first non-treated area NEA formed on an object-side surface of the lens and including a first step portion E11 may be different from a width G2 of a second non-treated area NEA formed on an image-side surface of the lens and including a second step portion E21 and a third step portion E22. In the example illustrated in FIG. **60**, G1 is greater than G2.

[0315] The width G1 of the first non-treated area NEA includes the first step portion E11, the second step portion E21, and the third step portion E22 when viewed in an optical axis direction, and the width G2 of the second non-treated area NEA includes the second step portion E21 and the third step portion E22 but not the first step portion E11 when viewed in the optical axis direction. A distance G4 from the outer end of the rib to the second step portion E21 is smaller than a distance G3 from the outer end of the rib to the first step portion E11. Also, a distance G5 from the outer end of the rib to the third step portion E22 is smaller than the distance G3 from the outer end of the rib to the first step portion E11.

[0316] The positions at which the non-treated areas NEA and the step portions E11, E21, and E22 are formed as described above may be advantageous for measuring a concentricity of the lens.

[0317] The examples described above enable the optical imaging system to be miniaturized and aberrations to be easily corrected.

[0318] While this disclosure includes specific examples, it will be apparent after an understanding of the disclosure of this application that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

Claims

1. An optical imaging system comprising: a first lens comprising a refractive power; a second lens comprising negative refractive power; a third lens comprising a refractive power; a fourth lens comprising a refractive power; a fifth lens comprising a refractive power; a sixth lens comprising a refractive power; and a seventh lens comprising a refractive power, wherein the first to seventh lenses are disposed in order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane, wherein absolute value of a radius of curvature of an image-side surface of the second lens is greater than an absolute value of a radius of curvature of an object-side surface of the first lens, wherein an absolute value of a radius of curvature of the object-side surface of the fourth lens is greater than an absolute value of a radius of curvature of an object-side surface of the second lens, wherein a thickness of the seventh lens is greater than a thickness of the first lens, and wherein $0.01<R1/R4<1.3$ and $0.05<R1/R6<0.9$, where R1 is a radius of curvature of the object-side surface of the first lens, R4 is a radius of curvature of the image-side surface of the second lens, and R6 is a radius of curvature of an image-side surface of the third lens.
2. The optical imaging system of claim 1, wherein the first lens has a convex object-side surface.
3. The optical imaging system of claim 1, wherein the second lens has a convex object-side surface.
4. The optical imaging system of claim 1, wherein the third lens has a convex object-side surface.
5. The optical imaging system of claim 1, wherein the fourth lens has a convex object-side surface.
6. The optical imaging system of claim 1, wherein the sixth lens has a convex object-side surface.
7. The optical imaging system of claim 1, wherein the seventh lens has a concave image-side surface.
8. An optical imaging system comprising: a first lens comprising a refractive power; a second lens comprising a refractive power; a third lens comprising a refractive power; a fourth lens comprising a convex object-side surface; a fifth lens comprising a convex object-side surface; a sixth lens comprising a refractive power; and a seventh lens comprising a refractive power, wherein the first to seventh lenses are disposed in order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane, wherein absolute value of a radius of curvature of an image-side surface of the second lens is greater than an absolute value of a radius of curvature of an object-side surface of the first lens, wherein an absolute value of a radius of curvature of the object-side surface of the fourth lens is greater than an absolute value of a radius of curvature of an object-side surface of the second lens, wherein a thickness of the seventh lens is greater than a thickness of the first lens, and wherein $0.01<R1/R4<1.3$ and $0.05<R1/R6<0.9$, where R1 is a radius of curvature of the object-side surface of the first lens, R4 is a radius of curvature of the image-side surface of the second lens, and R6 is a radius of curvature of an image-side surface of the third lens.
9. The optical imaging system of claim 8, wherein the first lens has a convex object-side surface.
10. The optical imaging system of claim 8, wherein the second lens has a convex object-side surface.
11. The optical imaging system of claim 8, wherein the third lens has a convex object-side surface.
12. The optical imaging system of claim 8, wherein the fifth lens has a concave image-side surface.
13. The optical imaging system of claim 8, wherein the sixth lens has a convex object-side surface.

14. The optical imaging system of claim 8, wherein the seventh lens has a concave image-side surface.